CHUCK Operation Manual

HF Propagation Link Reliability Forecasting System

Coverage & HF Usability Calculation Kit

4th edition 30 October 2026
Covers CHUCK v1.0 (screens as of 30 October 2026)

This manual explains how to use the CHUCK web screen to set up an HF link and check the 24-hour propagation, the link map, reports, area coverage, multiple RX sites, the station mesh, the 12-month forecast, the required transmit power and site candidates. Start with a calculation between fixed stations, then set up mobile stations, external observations and AI import of antenna documents as needed.

Your first calculation

1. Start with Docker and open http://localhost:18082 (for the public version, simply open the website).

2. On the right, choose one TX and one RX and specify their antennas.

3. Under “Forecast & space weather”, check the year and month, SSN, frequencies and emission class.

4. Press “Calculate 24-hour forecast →” and check the heatmap, the link map and the hourly details.

5. If needed, put everything on one page with “Print report”, or save CSV and JSON.

Getting this manual

You can always download the latest manual in the current screen language with “Manual (PDF)” at the top of the screen (Chapter 1).

Main changes in the 4th edition

This edition covers the outline that now marks only the selected cell in heatmaps (the whole time column when no frequency is selected) and the TX/RX site names in results, which are now shown in the screen language. All figures were retaken in all 8 languages. The manual is distributed as a PDF for each of the 8 screen languages (Japanese, English, French, Spanish, Italian, Portuguese, German and Korean).

> The figures show example operations on the English screen. Yokosuka, Guam and other places are approximate city positions and do not indicate real communication facilities. The results are examples calculated with the real VOACAP engine (November 2026, SSN 100 entered manually). External observations are shown before download, and the OCR results and model names are illustrative samples. Times are mainly shown in UTC; Japan time is UTC plus 9 hours.

1 Downloading the manual

The manual download button is at the top right of the screen, to the left of “What’s new”.

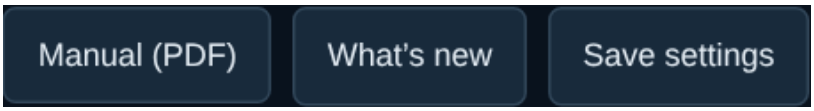


Figure 1 Manual download button

1. Choose the language you want to read with Language at the top right.

2. Press “Manual (PDF)” to save the illustrated PDF in the selected language. Use it for reading and printing.

3. Where the file is saved depends on your browser’s download settings.

> The manual is available in the same 8 languages as the screen, and its figures show the screen in each language. The button is available in both the public and the local versions. The version downloaded is the one bundled with the app at that time.

2 Starting and stopping

Start Docker Desktop and open a terminal in the repository folder. No AI API key is needed for normal propagation calculations. If you use the public version (a website that needs no installation), you can skip this chapter.

Purpose	Default address
Main operation screen	http://localhost:18082
Gateway administration screen	http://localhost:18083
Address between containers	http://position-gateway:8001

First start

```
git clone https://github.com/UeEmon/CHUCK-public-.git
cd CHUCK-public-
docker compose config --quiet
docker compose up -d --build
```

The first build needs an internet connection to fetch the base images and other components. After the build, calculations with a manual SSN and the map display work offline. On Mac/Linux you can also start with sh Start.sh, and in Windows PowerShell with ./Start.ps1.

Updating and restarting

```
git pull --ff-only
docker compose config --quiet
docker compose up -d --build
```

Stopping and checking logs

```
docker compose down
docker compose logs --tail=100 -f
```

Press Ctrl+C to leave the log display. This alone does not stop the containers.

Saved data

The calculation history, locations and antennas, imported source documents, saved AI settings and scheduled recalculations are stored in hf-data, and the station settings and latest positions in gateway-data. They are kept across normal stops and rebuilds. docker compose down -v deletes both persistent volumes, so do not use it for a normal shutdown.

To change the port, set HF_PORT=18084 or similar in .env. 18083 is the default port of the gateway. Do not set the same published port as the main app.

3 Screen layout and link map

The link map and results are on the left, and the TX and RX locations on the right. The tabs for results and settings are below the map.

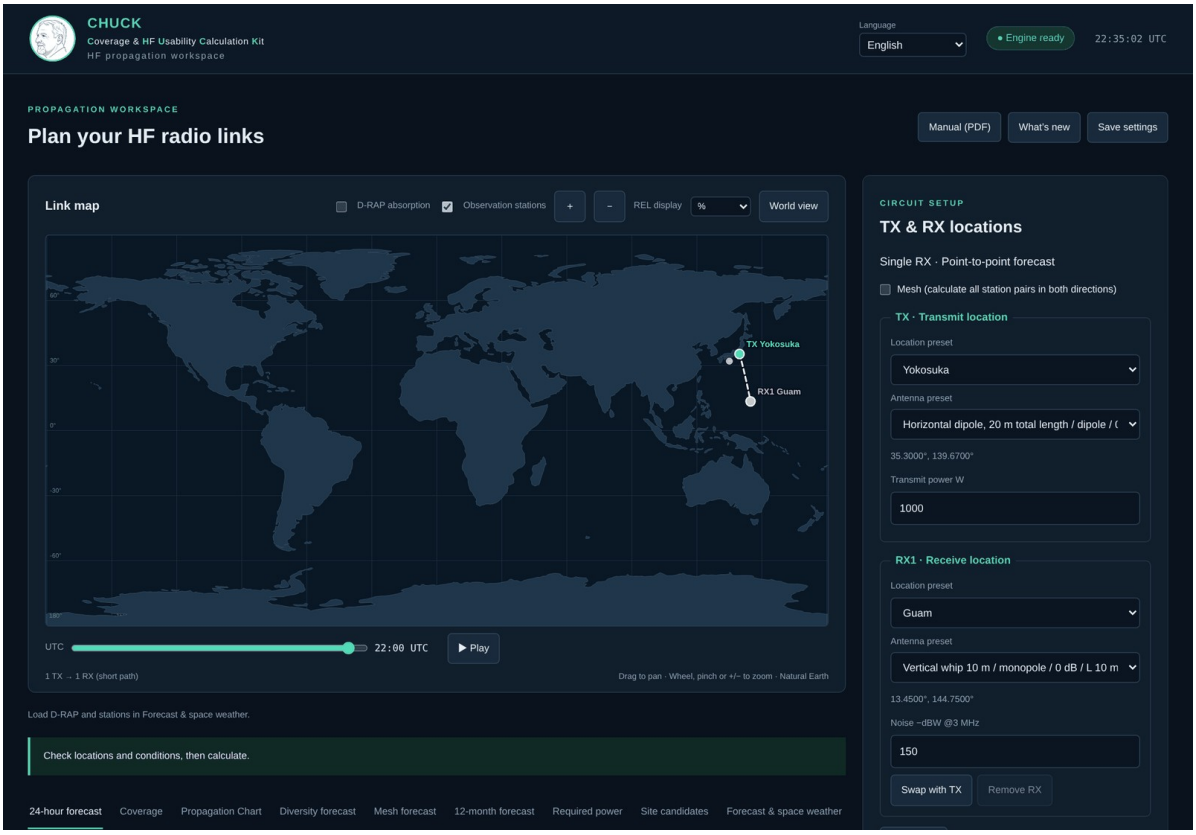


Figure 2 Initial screen of the main app

1. Under “TX & RX locations” on the right, specify TX, RX and antennas.

2. Drag the map to move the centre of the view. You can also move across longitude 180°.

3. Zoom in and out with the mouse wheel, by pinching on a touch screen or with “＋” and “－” at the top right of the map, and return to the initial view with “World view”.

4. Use “D-RAP absorption” and “Observation stations” to toggle the overlays of external observations (Chapter 21).

5. Use “REL display” to switch REL on the map and in tables between % and 4 grades (Chapter 9).

6. Use the tabs below the map to switch between results, conditions, presets, mobile stations and history.

> Left-clicking or right-clicking the map does not move TX/RX. Change locations by selecting presets. The map shows the current input, while results keep the conditions of the last calculation, so recalculate after any change.
>
> You can choose from 8 languages with Language at the top right. Built-in location names (Tokyo, Guam and so on) are also shown in the selected language in the mesh matrix, the diversity tables, the Propagation Chart site selector and reports. Station names that users registered or renamed, antenna names and entered values

are not changed. “What’s new” shows the history of added features (Chapter 29). “Save settings” saves the input conditions and the TX/RX selection in that browser.

4 Setting TX and RX locations

There is one TX and 1 to 8 RX. First use only one RX and check the point-to-point forecast. For the mesh, see Chapter 15.

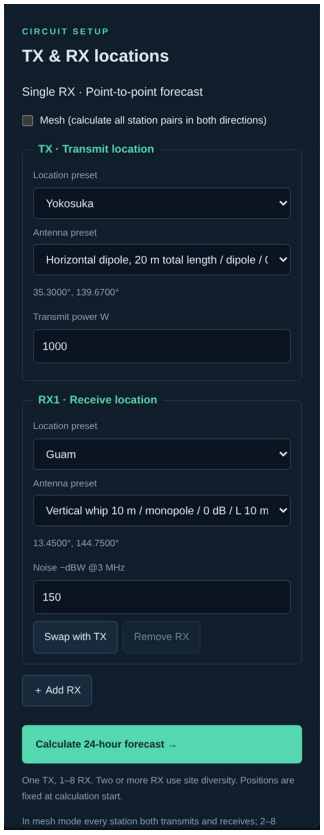


Figure 3 TX and RX settings

1. For TX, choose the “Location preset” and “Antenna preset”, and enter the transmit power in W.

2. Choose RX1 in the same way and enter the noise at the receive site. 150 means −150 dBW at 3 MHz.

3. “Swap with TX” exchanges the location, antenna and other settings between that RX and TX. Other RX are kept.

4. To add RX, press “＋ Add RX”. With two or more RX, the calculation becomes a diversity forecast.

5. Remove an RX you do not need with “Remove RX”. The last one cannot be removed.

To register new coordinates, open “Register / edit locations”; to edit antennas, open “Register / edit antennas”. Antennas marked as reference only cannot be selected for calculations.

5 Entering forecast conditions

Open the “Forecast & space weather” tab. The example screen uses November 2026, SSN 100 (manual input) and 10 frequencies from 2 to 20 MHz.

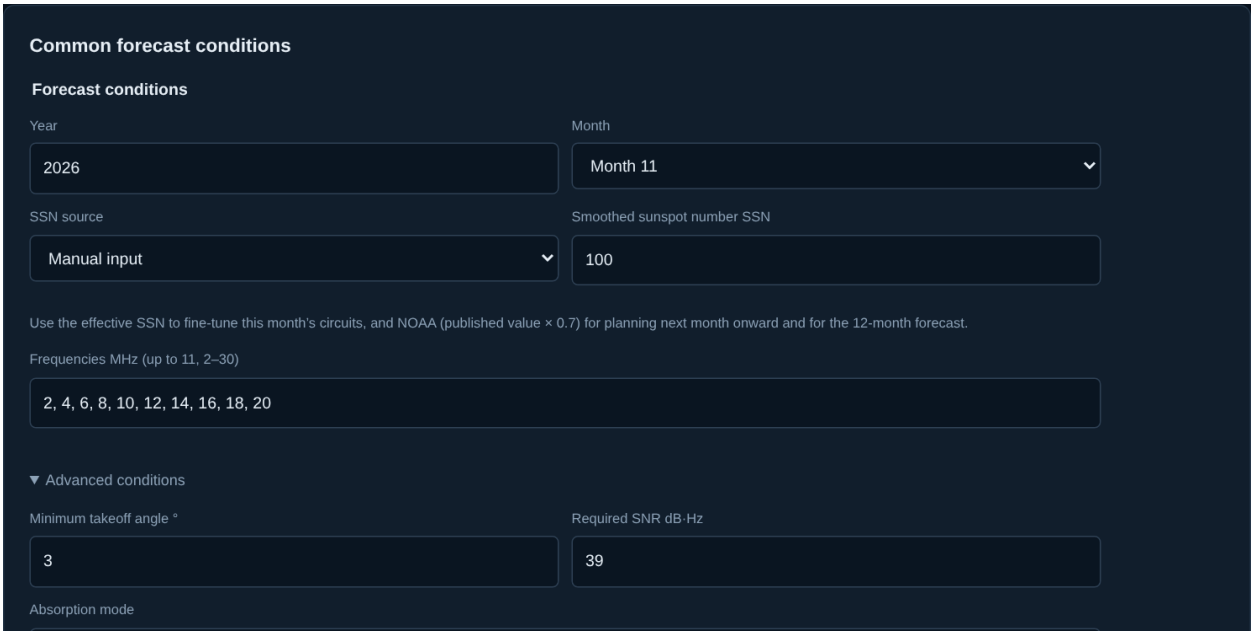


Figure 4 Common forecast conditions (top)

Item	Input and check
Year and month	Select the year and month to forecast. Hourly results are in UTC.
SSN source	Choose NOAA (the smoothed SSN for the month is fetched at calculation time and multiplied by 0.7), manual input (0 to 250) or effective SSN (calculated for the current month from observed foF2).
Frequencies MHz	Up to 11 values from 2 to 30 MHz, separated by commas or spaces.
Advanced conditions	Check the minimum take-off angle, required SNR and absorption mode.
REL display thresholds	The target REL and the boundaries of the 4-grade display (Chapter 7).



Figure 5 Advanced conditions (minimum take-off angle, required SNR, absorption mode)

1. To start, set “SSN source” to manual input and check the conditions.

2. Use effective SSN when refining links for the current month, and NOAA for planning next month or later and for the 12-month forecast. With NOAA, 0.7 times the published value is used in the calculation, and the results record both the published and the converted value.

3. The absorption mode under “Advanced conditions” defaults to a (the same as VOACAP Online Marine). The previous values can be reproduced with W.

4. Press “Calculate 24-hour forecast →” on the right. Wait until the calculation finishes.

Changing conditions alone does not update existing results. Recalculate after a change.

6 Emission class and polarization

Specify them in “Emission class & reception quality” and “Polarization & additional loss” in the same conditions tab.

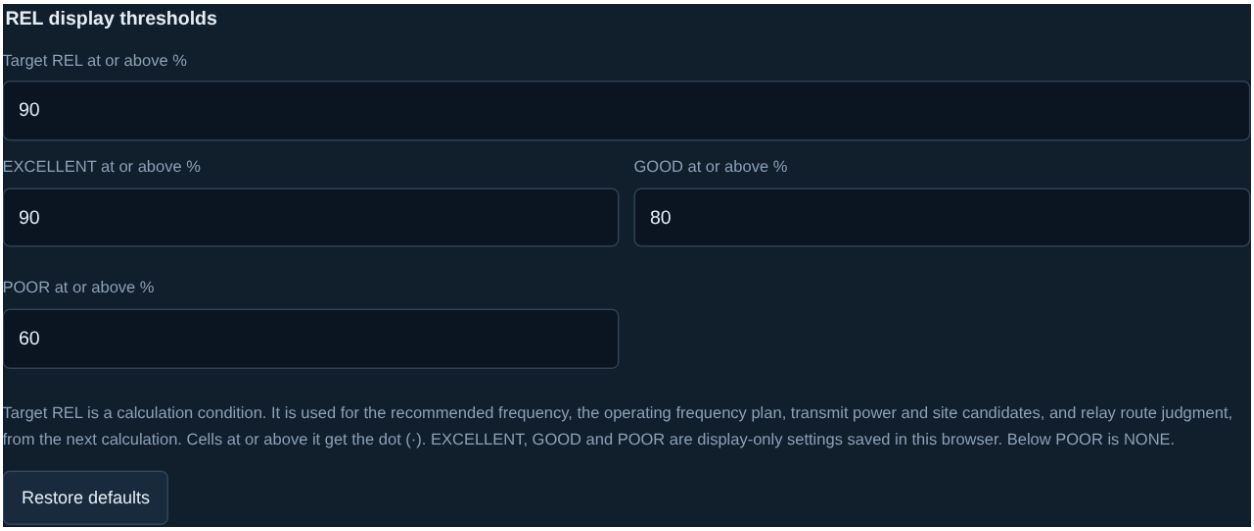


Figure 6 Emission class, reception quality and polarization

1. Choose the emission class (A1A, J3E, A3E, F1B, G1D and so on). The required SNR values of 19 for A1A, 39 for J3E and 48 dB·Hz for A3E are examples (the same as VOACAP Online Marine). Change them as needed.

2. Enter the occupied bandwidth in emission designator notation (3K00, 2K40, 500H and so on).

3. To enter the in-band SNR, choose “Convert in-band SNR and occupied bandwidth” and specify the required SNR. Example: 3K00 and 10 dB convert to about 44.77 dB·Hz.

4. For FSK, PSK and similar modes, specify the required SNR that matches the modem, coding, speed and target BER.

5. Choose the polarization for TX and each RX, and if needed specify an additional polarization loss common to all paths (0 to 30 dB).

> The polarization name is recorded as a condition. Changing the name alone does not switch the antenna pattern, and polarization mismatch and Faraday rotation are not calculated automatically. The additional loss is applied as an equivalent power reduction. Do not count losses twice that are already included in the antenna gain or cable loss. Demodulation of waveforms, delay spread and the BER itself are not calculated.

7 Target REL and REL display thresholds

Under “REL display thresholds”, set the target REL used as the pass criterion and the boundaries of the 4-grade display.

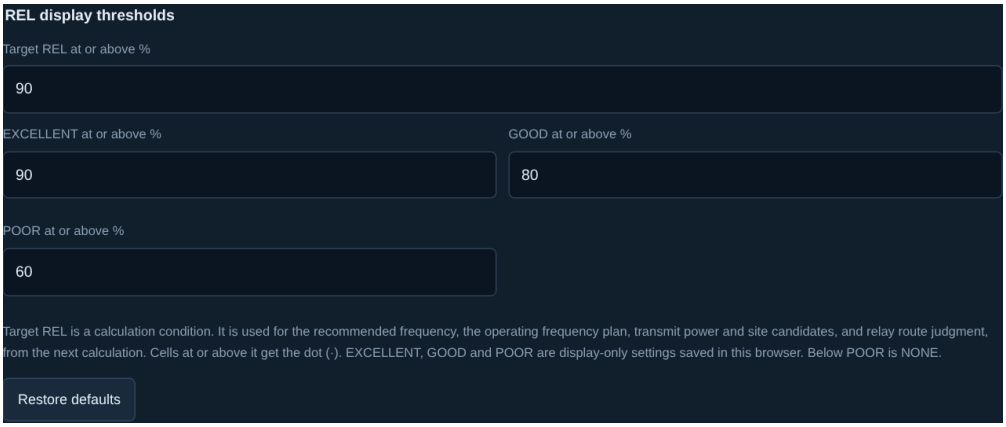


Figure 7 REL display thresholds

1. Enter “Target REL at or above %”. It is used for the recommended frequency, the mark (·) in tables, the operating frequency plan and the judgements of the mesh, site candidates and required power.
2. Change the boundaries of the 4-grade display (EXCELLENT, GOOD, POOR, NONE) as needed.
3. “Restore defaults” returns the boundaries to their initial values.

The mark (·) in heatmap cells and elsewhere shows that the target REL is met, and it changes along with the target REL. The target REL is saved with the results as a calculation condition, so recalculate after changing it. Changes to the boundaries only affect the display.

8 Checking the 24-hour forecast

With one RX, the results are shown under “24-hour forecast”.



Figure 8 Link distance, bearing, recommended frequency and reliability heatmap (with time slider)

1. Select the cell where a frequency row and a UTC column cross. The colour and the number show REL, and the mark (·) means the target REL is met. The selected cell is outlined and the headings of its frequency and time are highlighted. The row for that frequency is also highlighted in the hourly details.

2. Selecting the same cell again clears the frequency selection, and the whole column of the selected time is outlined.

3. Switch the time with the UTC slider above the heatmap and “▶Play”. The hourly details and the link map switch to the same time.

4. The top row shows the link distance, the bearing from the transmit site and the recommended frequency at the selected time.

5. Check REL, SNR, signal strength and take-off angle in “Hourly details” or by selecting a UTC.

6. The recommended frequency is chosen from the input frequencies that meet the target REL. “Below target” means there is no such frequency.

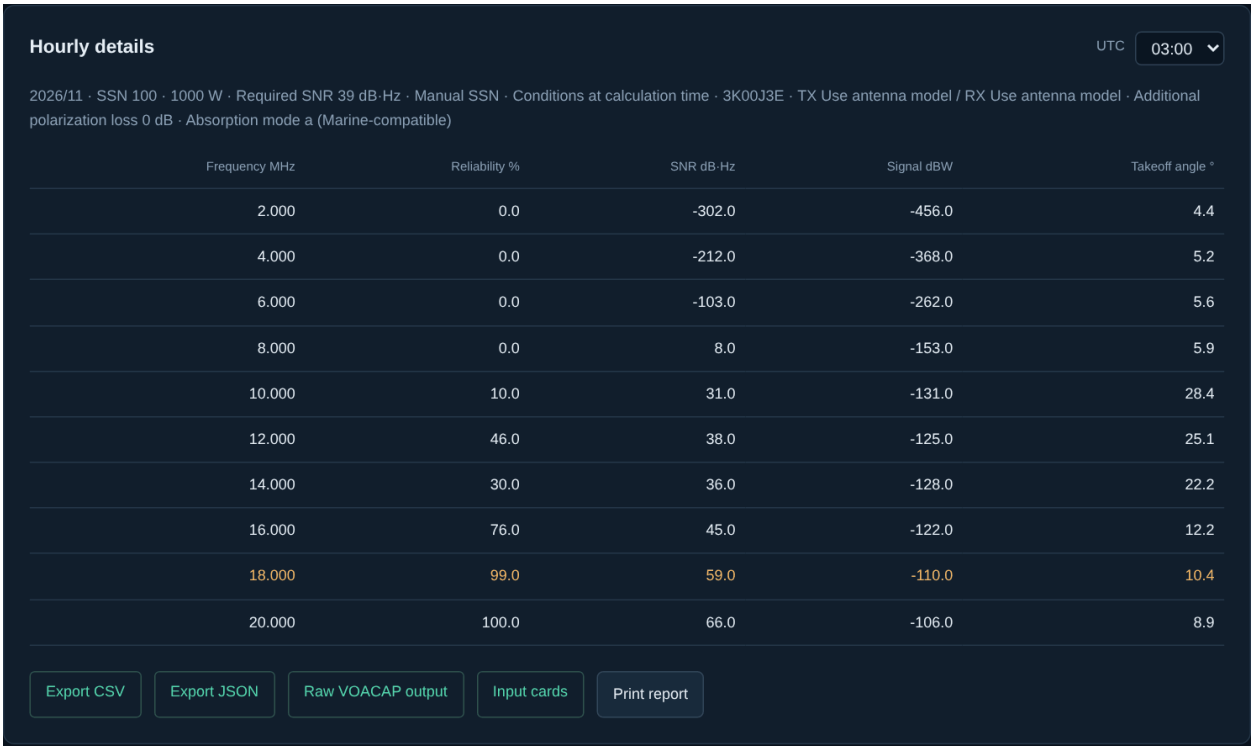


Figure 9 Hourly details and output

From the output area you can use “Print report” (Chapter 11) and save CSV, JSON, the raw VOACAP output and the input card.

REL is the model’s link reliability: the fraction of time within the month that meets the required SNR. It is not the instantaneous probability of connecting. SNR is in dB·Hz (normalized to 1 Hz); do not compare it directly with the in-band SNR of a receiver.

9 Link map time slider and REL display

In the 24-hour forecast, diversity forecast and mesh forecast tabs, a UTC slider and play/stop buttons are shown below the link map.

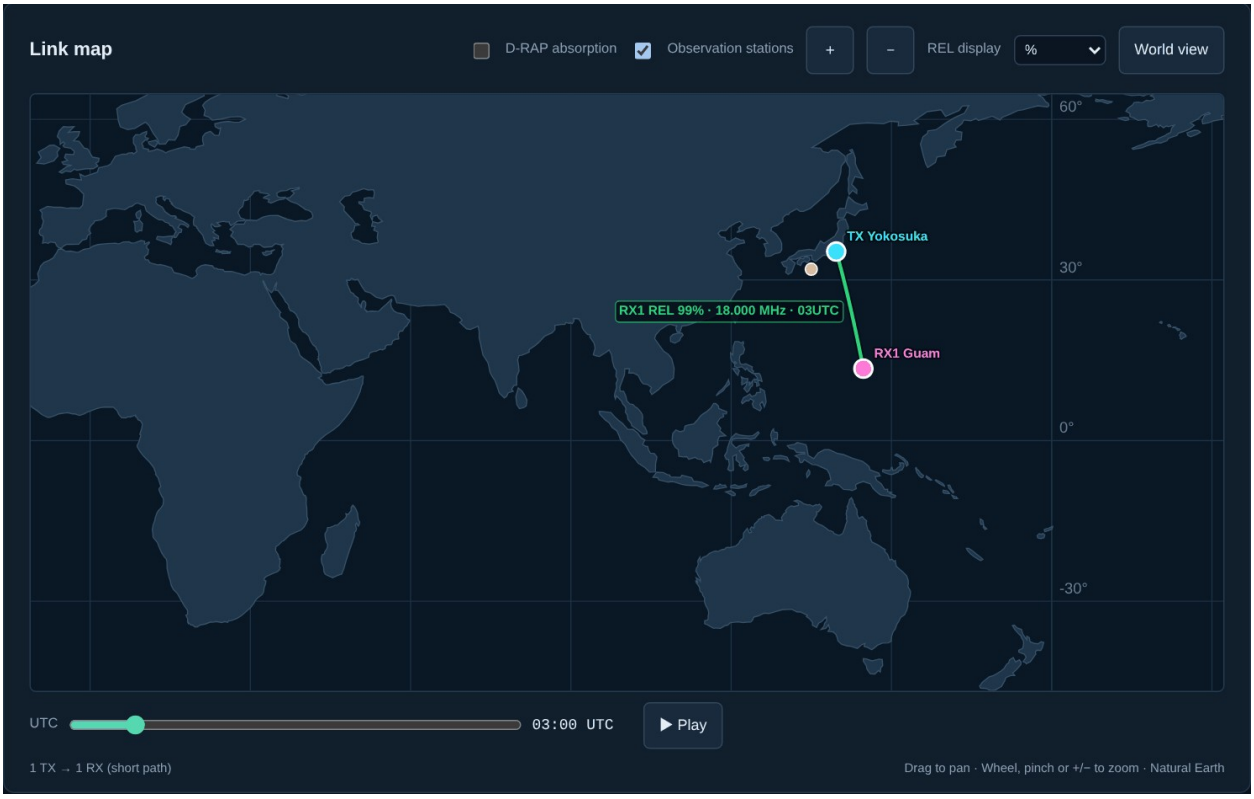


Figure 10 Time slider and paths coloured by REL

1. Moving the slider switches the colours of the paths on the map and the REL labels on the paths (REL, frequency, UTC) to that time.

2. “▶Play” advances automatically one hour at a time, and “■Stop” stops it.

3. Selecting a time with a heatmap cell or with the slider above the heatmap also moves the map to that time.

4. While a frequency row is selected in the heatmap, the path colours and REL are also shown for that frequency. Selecting the same cell again clears the selection and returns to the best-frequency display.

REL display (% / 4 grades)

When “REL display” at the top right of the map is set to “4 grades”, the map, heatmaps and mesh table show the grades EXCELLENT, GOOD, POOR and NONE. The colours of the % display use the same hues as the 4-grade display, so both read with the same colours.

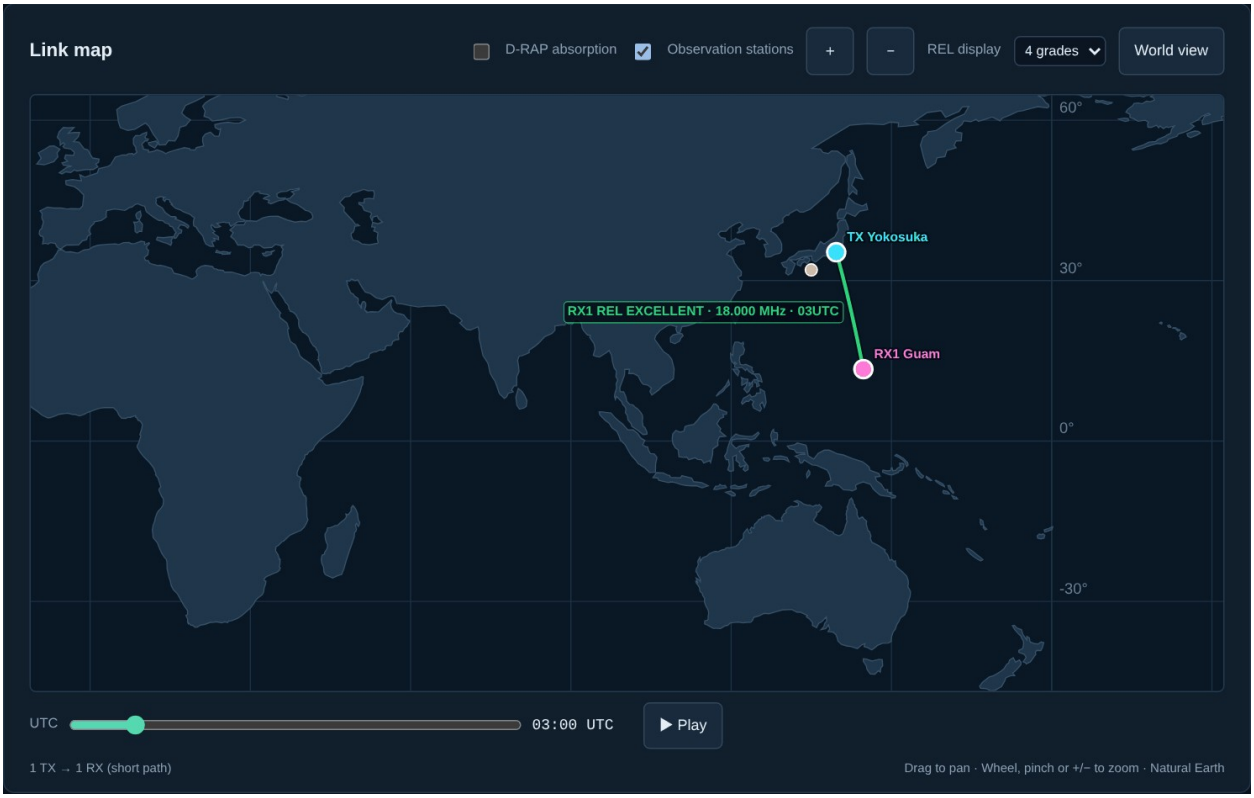


Figure 11 Link map in the 4-grade display



Figure 12 Heatmap in the 4-grade display

When no frequency is selected, the path colour shows the best REL at the selected time. The grade boundaries can be changed as described in Chapter 7.

10 24-hour operating frequency plan

Below the 24-hour forecast, a plan table shows which frequency to use in each time slot. The calculated frequencies (up to 11) are treated as the assigned frequencies, and the plan chooses among them.

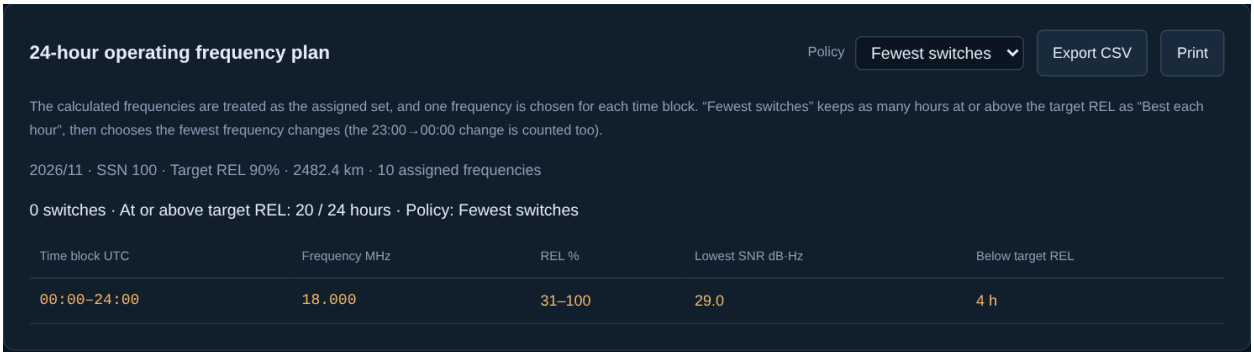


Figure 13 24-hour operating frequency plan

1. Check the time slot, frequency, REL (minimum to maximum), minimum SNR and the number of hours below the target REL.

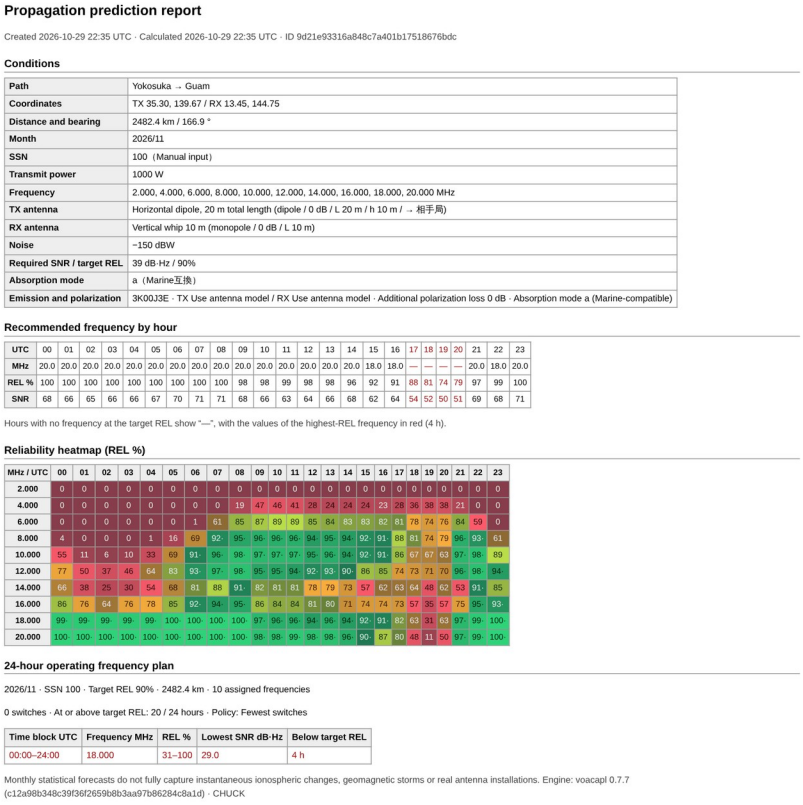
2. Switch the “Policy”. “Fewest switches” (default) keeps as many hours at or above the target REL as “Best each hour” and then minimizes the number of switches. “Best each hour” chooses the frequency with the highest REL at each UTC.

3. “Export CSV” saves one row per hour, and “Print” prints only the plan table.

> The plan assumes the same daily plan repeats every day, so a switch from 23:00 to 0:00 also counts as one switch. In hours when no frequency reaches the target REL, the best frequency is shown and the hour is counted as “Below target REL”.

11 Printing reports

“Print report” puts the conditions and the tables and charts shown on one white printable page. It is available in the 24-hour forecast and also in the Coverage, Propagation Chart, Diversity forecast, Mesh forecast, 12-month forecast, Required power and Site candidates tabs.



Propagation prediction report

Created 2026-10-29 22:35 UTC · Calculated 2026-10-29 22:35 UTC · ID 9d21e93316a848c7a401b17518676bdc

Conditions

Path	Yokosuka → Guam
Coordinates	TX 35.30, 139.67 / RX 13.45, 144.75
Distance and bearing	2482.4 km / 166.9 °
Month	2026/11
SSN	100（Manual input）
Transmit power	1000 W
Frequency	2.000, 4.000, 6.000, 8.000, 10.000, 12.000, 14.000, 16.000, 18.000, 20.000 MHz
TX antenna	Horizontal dipole, 20 m total length (dipole / 0 dB / L 20 m / h 10 m / → 相手局)
RX antenna	Vertical whip 10 m (monopole / 0 dB / L 10 m)
Noise	−150 dBW
Required SNR / target REL	39 dB·Hz / 90%
Absorption mode	a（Marine互換）
Emission and polarization	3K00J3E · TX Use antenna model / RX Use antenna model · Additional polarization loss 0 dB · Absorption mode a (Marine-compatible)

Recommended frequency by hour

UTC	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
MHz	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	18.0	18.0	—	—	—	—	20.0	18.0	20.0
REL %	100	100	100	100	100	100	100	100	100	98	98	99	98	98	96	92	91	88	81	74	79	97	99	100
SNR	68	66	65	66	66	67	70	71	71	68	66	63	64	66	68	62	64	54	52	50	51	69	68	71

Hours with no frequency at the target REL show "—", with the values of the highest-REL frequency in red (4 h).

Reliability heatmap (REL %)

MHz / UTC	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
2.000	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4.000	0	0	0	0	0	0	0	0	19	47	46	41	28	24	24	24	23	28	36	38	38	21	0	0
6.000	0	0	0	0	0	0	1	61	85	87	89	89	85	84	83	83	82	81	78	74	76	84	59	0
8.000	4	0	0	0	1	16	69	92	95	96	96	96	94	95	94	92	91	88	81	74	79	96	93	61
10.000	55	11	6	10	33	69	91	96	98	97	97	97	95	96	94	92	91	86	67	67	63	97	98	89
12.000	77	50	37	46	64	83	93	97	98	95	95	94	92	93	90	86	85	74	73	71	70	96	98	94
14.000	66	38	25	30	54	68	81	88	91	82	81	81	78	79	73	57	62	63	64	48	62	53	91	85
16.000	86	76	64	76	78	85	92	94	95	86	84	84	81	80	71	74	74	73	57	35	57	75	95	93
18.000	99	99	99	99	99	99	100	100	100	100	97	96	96	94	96	94	92	91	82	63	31	63	97	99	100
20.000	100	100	100	100	100	100	100	100	100	98	98	99	98	98	96	90	87	80	48	11	50	97	99	100

24-hour operating frequency plan

2026/11 · SSN 100 · Target REL 90% · 2482.4 km · 10 assigned frequencies

0 switches · At or above target REL: 20 / 24 hours · Policy: Fewest switches

Time block UTC	Frequency MHz	REL %	Lowest SNR dB·Hz	Below target REL
00:00–24:00	18.000	31–100	29.0	4 h

Monthly statistical forecasts do not fully capture instantaneous ionospheric changes, geomagnetic storms or real antenna installations. Engine: voacapl 0.7.7 (c12a98b348c39f36f2659b8b3aa97b86284c8a1d) · CHUCK

Figure 14 24-hour forecast report (printable page)

1. After the calculation, press “Print report” in the output area of each tab (for the Propagation Chart, above the chart).

2. A separate window opens and the print dialog appears. If you need a PDF, choose “Save as PDF” as the destination. The layout assumes A4 landscape.

3. If pop-ups are blocked, allow pop-ups for this page in your browser.

Tab	Contents
24-hour forecast	Conditions, recommended frequency by hour, reliability heatmap, 24-hour operating frequency plan
Coverage	Frequency, UTC, number of points, and the link map with the grid points
Propagation Chart	Selected result, receiving station, display metric and chart
Diversity forecast	Table of the best site’s REL, site selection and theoretical range, details by site
Mesh forecast	TX × RX matrix (REL after relaying and the↻number of relays), table of the

	selected path and the relay route
12-month forecast	Month × UTC table, and the frequency × UTC table of the selected month
Required power	Table of minimum power, and table of the best REL for each power level
Site candidates	Table of comparison results

The colours and marks (·) in the tables are the same as on screen and reflect the target REL, the REL display thresholds and the % / 4-grade switch. A report shows the results at the time of calculation and does not indicate any frequency assignment or operating permission.

12 Using the Propagation Chart

In the “Propagation Chart” tab you can see trends by UTC and frequency. Run a point-to-point or diversity calculation first.



Figure 15 Propagation Chart by UTC and frequency

1. Under “Forecast shown”, choose the point-to-point forecast or the diversity forecast.
2. With multiple RX, select the “Received station” and check each path.
3. Switch the “Display metric” between REL and SNR.
4. Toggle the MUF display. Selecting a cell moves to the details for that time.
5. Save the chart with “Export SVG” and print with “Print report”.

Cells are discrete values of the calculated frequencies and UTC hours. There is no interpolation between frequencies. The MUF is the engine output for the selected path and is not a boundary that guarantees successful communication or permission to use a frequency.

13 Calculating area coverage

“Coverage” calculates a regional grid around the position of RX1 for a specified time and frequency.

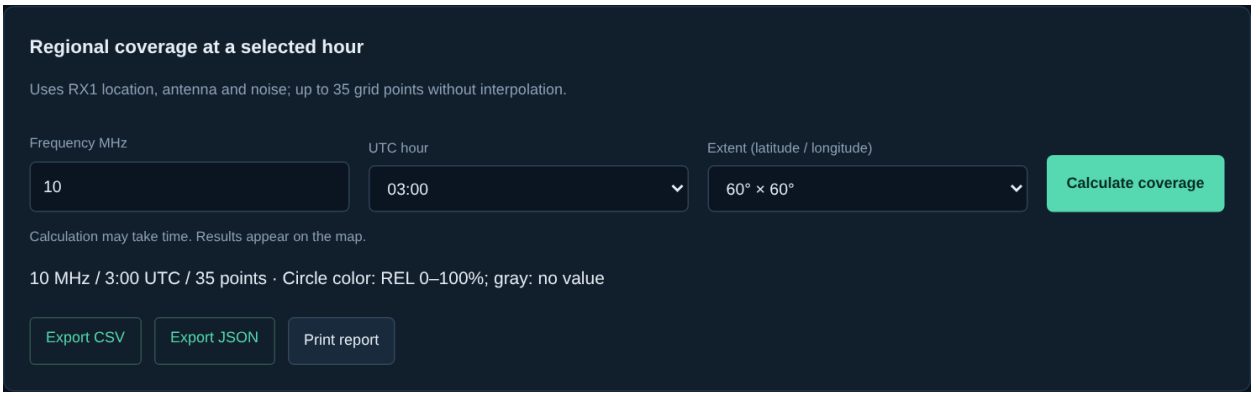


Figure 16 Specifying frequency, time and area



Figure 17 Coverage shown on the map

1. In the “Coverage” tab, choose the frequency, UTC time and the latitude and longitude spans.

2. Check the location, antenna and noise of RX1 and the common conditions. Even if you add RX2 or more, this calculation uses RX1.

3. Press “Calculate coverage” and wait until up to 35 points have been processed.

4. Check the colours of the circles on the map, and select a circle to show its location and REL. You can also save CSV/JSON and print a report.

Each circle is an independent point-to-point forecast. There is no interpolation between circles, and they do not estimate the boundary of a continuous area where communication is possible. Grey means no calculated value.

14 Diversity forecast with multiple RX

Set 2 to 8 RX on the right and press “Calculate 24-hour forecast →” to show the “Diversity forecast”.

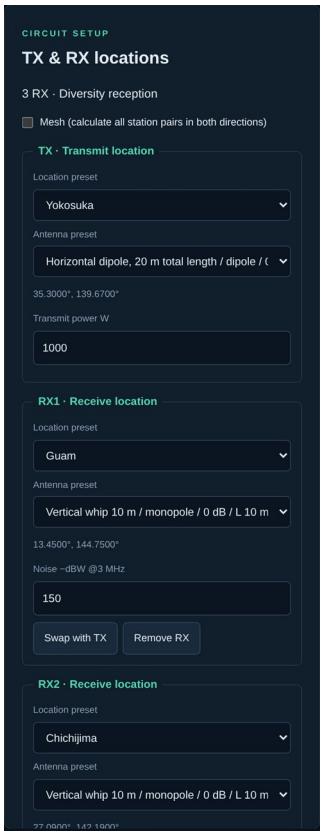


Figure 18 TX and RX locations with three RX



MHz / UTC	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
2.000	0	0	0	0	0	0	0	0	6	22	25	19	9	6	7	8	8	13	24	33	34	10	0	0
4.000	0	0	0	0	0	0	19	80	94	94	94	93	91	86	86	85	85	86	88	90	91	93	91	23
6.000	78	37	15	17	54	96	99	100	100	100	100	99	99	98	98	98	97	98	97	97	98	99	100	99
8.000	99	99	96	96	99	100	100	100	100	100	100	100	100	98	97	97	97	97	97	96	93	100	100	100
10.000	100	100	97	100	100	100	100	100	100	100	100	100	99	97	97	95	95	97	96	83	73	97	100	100
12.000	100	100	100	100	100	100	100	100	100	100	99	98	98	95	90	86	85	79	79	71	70	96	100	100
14.000	99	99	98	99	99	99	99	99	99	93	90	91	90	79	73	57	62	63	64	48	62	53	98	99
16.000	100	99	99	99	99	100	100	99	97	97	94	93	87	80	71	74	74	73	57	35	57	75	95	100
18.000	100	100	100	100	100	100	100	100	100	100	99	96	94	96	94	92	91	82	63	31	63	97	99	100
20.000	100	100	100	100	100	100	100	100	100	100	98	99	98	98	96	90	87	80	48	11	50	97	99	100

Figure 19 REL heatmap of the best site

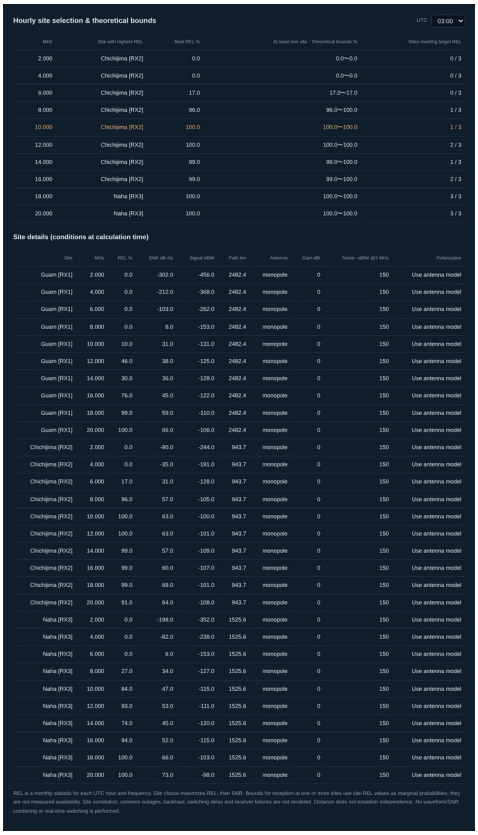


Figure 20 Site comparison and details by site

1. Set the location, antenna and noise for each RX.

2. Select a time with a heatmap cell, or with the UTC slider above the heatmap and “▶Play”, and check the best site for each frequency and the details by site. The selected cell is outlined; when no frequency is selected, the whole time column is outlined.

3. Compare the “Sites meeting target REL” and, if needed, switch RX in the Propagation Chart.

4. On the link map, the paths to each RX are coloured by REL according to the time slider.

5. Only when you need the reference value under the independence assumption, enable the corresponding checkbox in the conditions tab and recalculate.

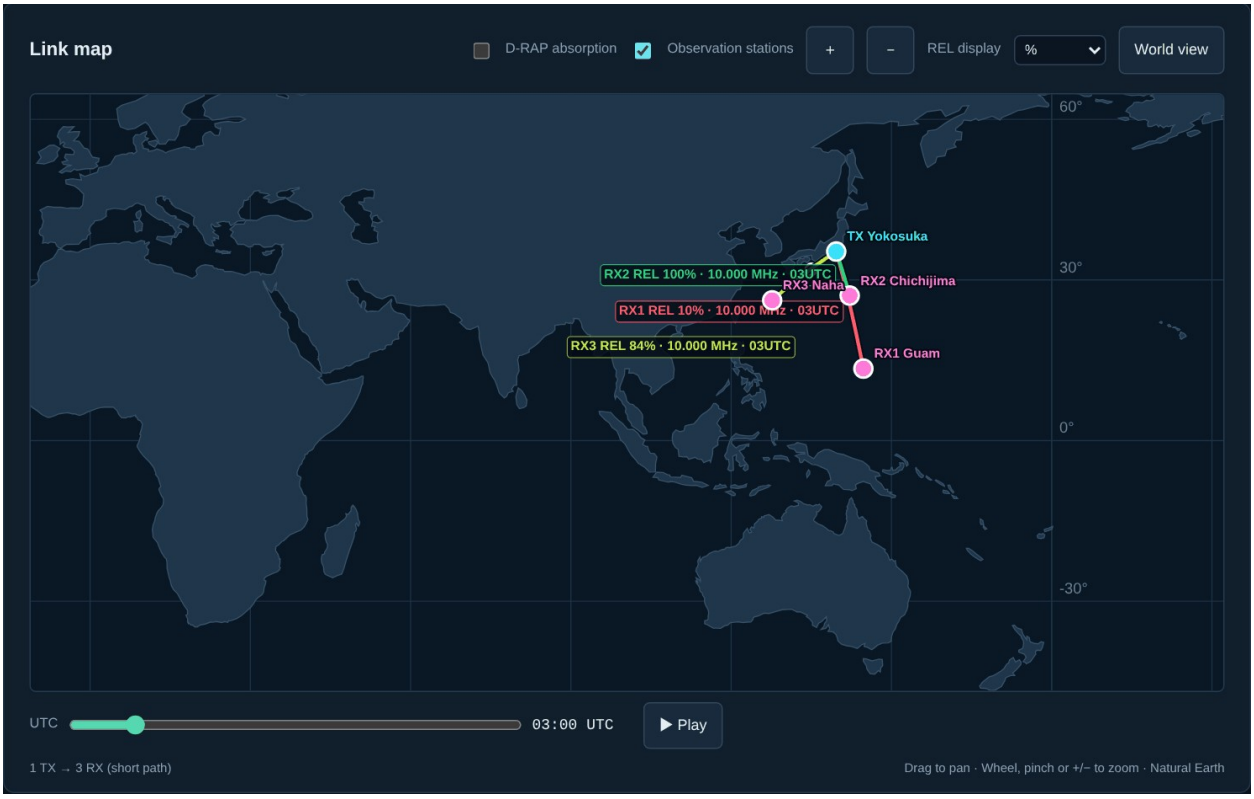


Figure 21 Link map of the diversity forecast

The heatmap shows the REL of the best site, not a combined reliability. In an example with REL of 80% and 60%, the theoretical range for reception at at least one site is 80 to 100%, and the reference value under the independence assumption is 92%. Correlation between sites is not evaluated. Waveform combining, actual receiver switching, and availability including common failures and IP relay links are not evaluated.

15 Station mesh forecast

For a set of 2 to 8 stations that each transmit and receive, the REL between all stations is shown as a TX (rows) × RX (columns) matrix. With N stations, N×(N−1) paths (56 paths for 8 stations) are calculated, one per direction.

1. Under “TX & RX locations” on the right, choose “Mesh (calculate all station pairs in both directions)”. The TX and RX cards change to “Station 1 to N”.

2. For each station, set the location, transmit antenna, receive antenna (the same as the transmit antenna by default), transmit power and noise. “+ Add station” allows up to 8 stations.

3. “Calculate 24-hour forecast →” moves to the “Mesh forecast” tab.

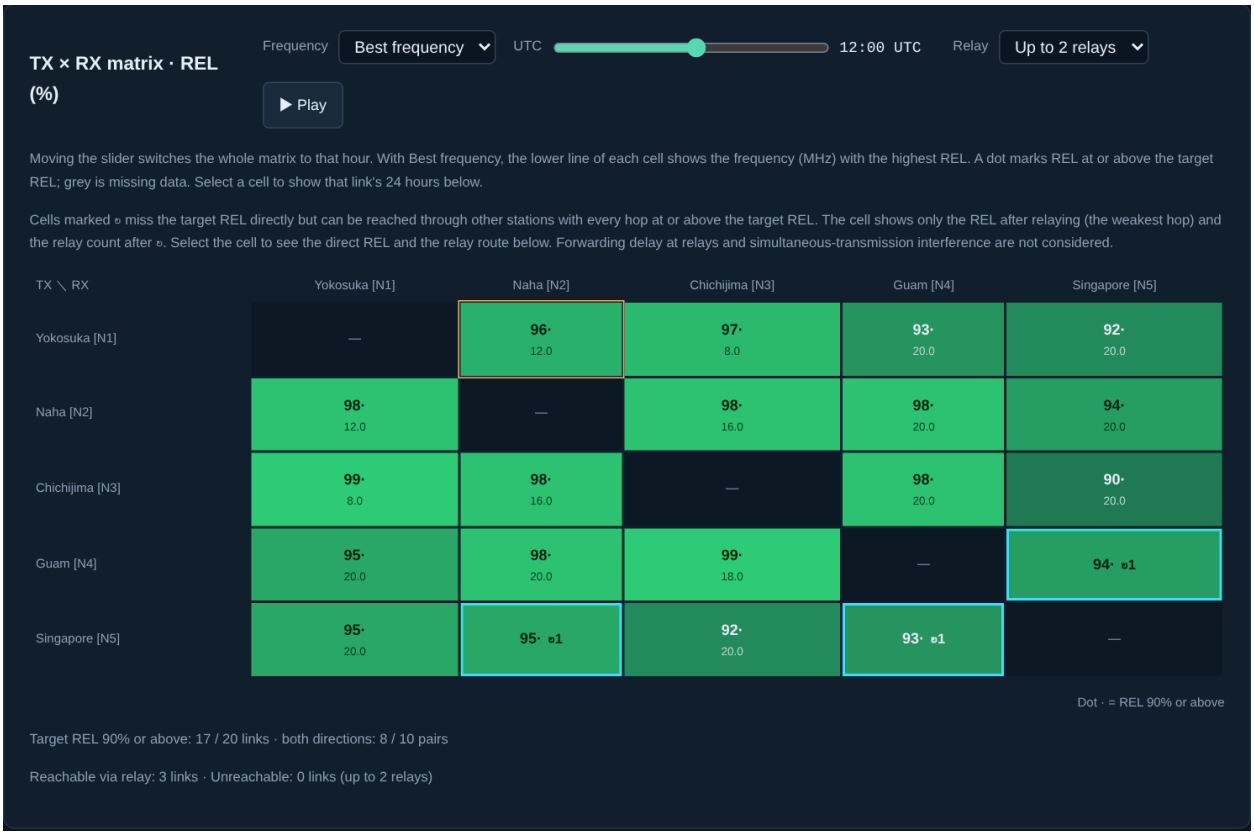


Figure 22 TX × RX matrix (paths reached by relaying are outlined and show the↻number of relays)

1. Use the UTC slider and “▶Play” to switch the time of the whole matrix. The time of the link map follows.

2. Specify a “Frequency” or choose “Best frequency”. With the best frequency, the frequency is shown in the lower line of each cell.

3. Choose “Relay” from “Direct only”, “Up to 1 relay” and “Up to 2 relays” (default). Paths that do not reach the target REL directly but do so by relaying are outlined and show the REL after relaying and the↻number of relays, with the REL without relaying below.

4. Selecting a cell shows the frequency × UTC REL and the relay route of that path in “Link detail · frequency × UTC” below. In the link detail, the cell for the selected time and frequency is outlined (the whole time column for the best frequency). “Show reverse direction” swaps TX and RX.

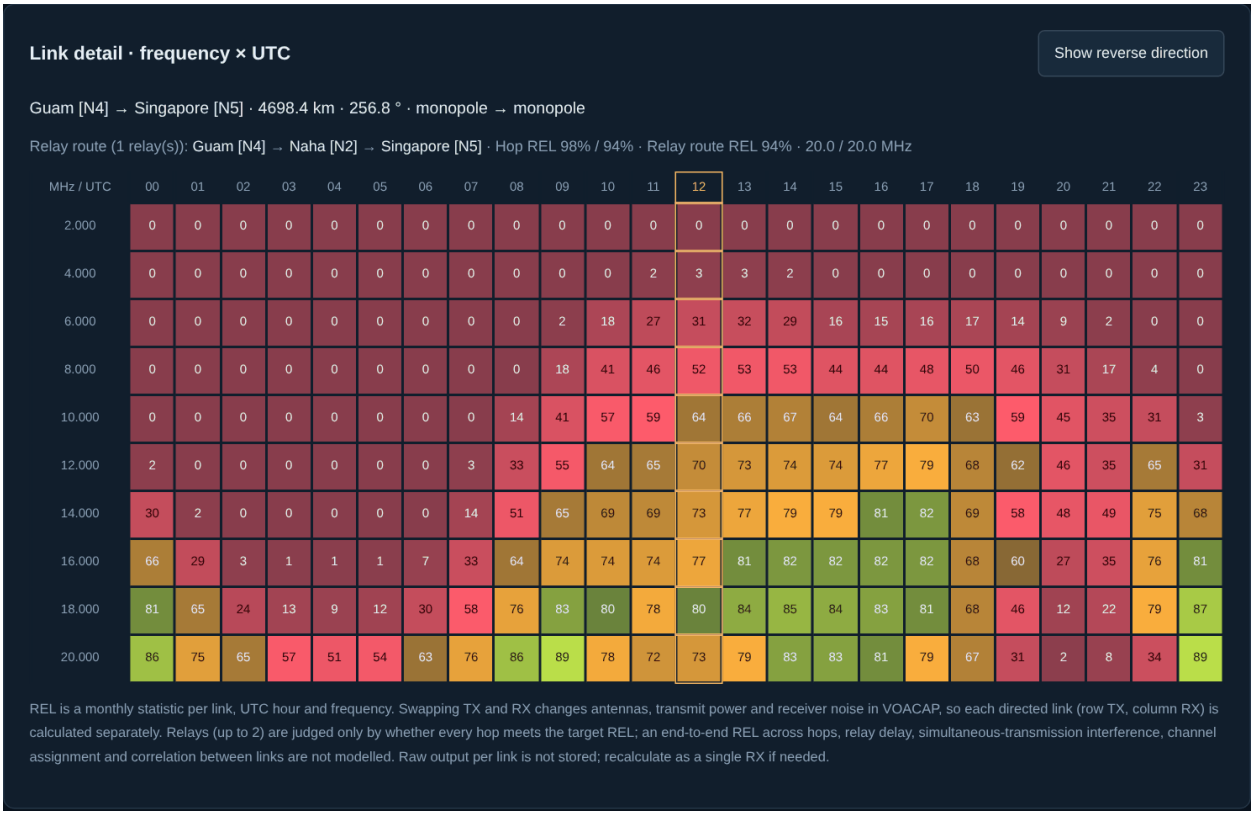


Figure 23 Link detail and relay route



Figure 24 Link map of the mesh forecast (two paths, one in each direction, for each pair of stations)

For each pair of stations, the link map draws the two paths A→B and B→A side by side, with arrows for the direction and colours and labels for the REL of each direction.

Relaying is judged only by whether every hop meets the target REL at the same UTC; an end-to-end REL obtained by multiplying the hop RELs is not shown. Store-and-forward delay at relay stations, transmit/receive switching, interference and channel assignment are not considered. Because VOACAP inputs are asymmetric between transmission and reception, the REL of A→B and B→A may differ.

16 12-month forecast

For one path (TX→RX1), 24-hour forecasts are calculated for 12 months starting from the year and month in the forecast conditions, and the month × UTC REL shows the seasonal changes.



Figure 25 12-month forecast (REL by month × UTC)

1. Choose TX and RX on the right. If there are several RX, RX1 is used. It cannot be used with the mesh.

2. In the “12-month forecast” tab, press “Calculate 12 months”. The engine runs 12 times in sequence.

3. Rows of the table are months and columns are UTC. Use “Frequency” to choose the best frequency or a specific frequency. The rightmost column is the number of hours at or above the target REL.

4. Selecting a cell shows the frequency × UTC of that month in “Month detail · frequency × UTC” below. You can save CSV/JSON and print a report.

> A manual SSN uses the same value for all 12 months. NOAA uses the smoothed value for each month, and months without a fresh value are shown as grey rows. Effective SSN cannot be used.

17 Required power

The transmit power is varied in 10 steps from 10 W to 10 kW, and the minimum power that reaches the target REL is shown for each time and frequency.



Figure 26 Minimum power (W) by frequency × UTC

1. Choose TX and RX1 on the right. The transmit power value on the right is not used in the calculation (it is marked in the table for reference).

2. In the “Required power” tab, press “Find required power”. The engine runs 10 times in sequence.

3. In the upper table, check the lowest step (W) that reaches the target REL. “>10k” means it is not reached even with 10 kW. The top row, “Minimum power”, shows the frequency and power that need the least power at that time.

4. In the lower table, check the best REL by hour for each power level and the number of hours at or above the target REL. The row with the same power as the current setting is outlined.



Best REL per power · power × UTC

W / UTC	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	At target REL (h)
10	83	77	76	77	79	83	88	89	88	74	71	66	66	70	70	62	60	48	23	15	23	73	76	89	0
20	90·	86	85	86	88	90·	93·	94·	93·	80	78	75	74	78	76	68	66	54	28	24	29	79	83	93·	6
50	96·	94·	93·	94·	95·	96·	97·	98·	97·	87	86	85	84	86	83	74	74	62	49	40	44	86	90·	97·	11
100	98·	97·	97·	97·	98·	98·	99·	99·	99·	91·	91·	90·	89	90·	87	79	79	67	58	52	55	90·	94·	99·	16
200	99·	99·	98·	99·	99·	99·	100·	100·	99·	94·	94·	94·	93·	94·	91·	84	83	74	65	59	63	93·	96·	99·	18
500	100·	100·	100·	100·	100·	100·	100·	100·	100·	97·	97·	97·	96·	97·	94·	89	88	83	75	68	73	96·	98·	100·	18
1000	100·	100·	100·	100·	100·	100·	100·	100·	100·	98·	98·	99·	98·	98·	96·	92·	91·	88	81	74	79	97·	99·	100·	20
2000	100·	100·	100·	100·	100·	100·	100·	100·	100·	99·	99·	99·	99·	99·	97·	95·	95·	92·	86	81	85	99·	100·	100·	21
5000	100·	100·	100·	100·	100·	100·	100·	100·	100·	100·	100·	100·	100·	100·	99·	98·	98·	96·	91·	88	90·	99·	100·	100·	23
10000	100·	100·	100·	100·	100·	100·	100·	100·	100·	100·	100·	100·	100·	100·	99·	99·	99·	97·	94·	91·	93·	100·	100·	100·	24

For each power, shows the highest REL at each hour. A dot marks the target REL or above. The last column is the number of hours at or above the target REL.

The power is varied in 1-2-5 steps from 10 W to 10 kW and the smallest step that reaches the target REL is shown. Powers between steps are not calculated. Conditions other than power (antennas, noise, required SNR, polarization loss) keep their values at calculation time. REL is a monthly statistic, not whether a contact succeeds at a given moment.

Figure 27 Best REL for each power level

Powers between steps are not calculated. “200” means the target is not reached with 100 W but is reached with 200 W. The permissible power of the transmitter, power supply and radio regulations is not considered.

18 Site candidates

Keeping the current TX (and RX1 for relaying) fixed, compare which registered location presets reach best as a receive site or a relay site.

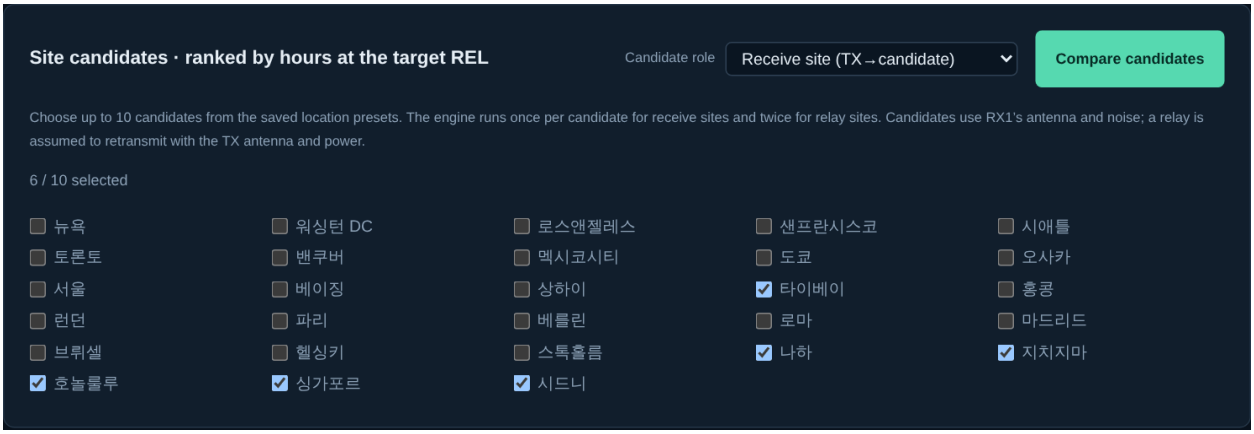


Figure 28 Candidate role and selecting candidate sites

1. Choose the “Candidate role”. Receive site (TX→candidate) runs the engine once per candidate, and relay site (TX→candidate→RX1) twice.

2. Select up to 10 candidates from the location presets. The TX and RX1 locations and mobile stations are not offered as candidates.

3. Press “Compare candidates” to sort them by the number of hours at or above the target REL (ties by higher average REL).

4. The 24 boxes in each row are the REL of the best frequency at each hour. For relaying, the lower of the two hops is shown.

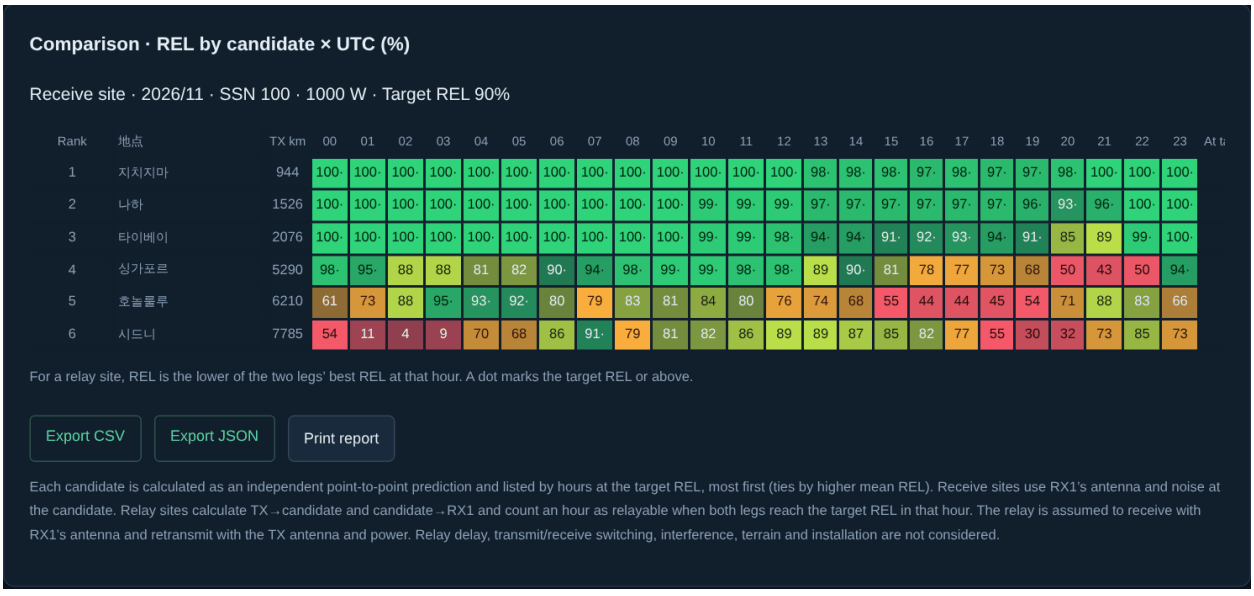


Figure 29 Comparison results (REL by candidate × UTC)

Candidate sites use the antenna and noise of RX1. Differences in terrain, local noise environment and installation conditions are not considered.

19 Registering and editing location presets

Location coordinates and names are kept together in “Location preset”. Enter southern latitudes and western longitudes as negative numbers.

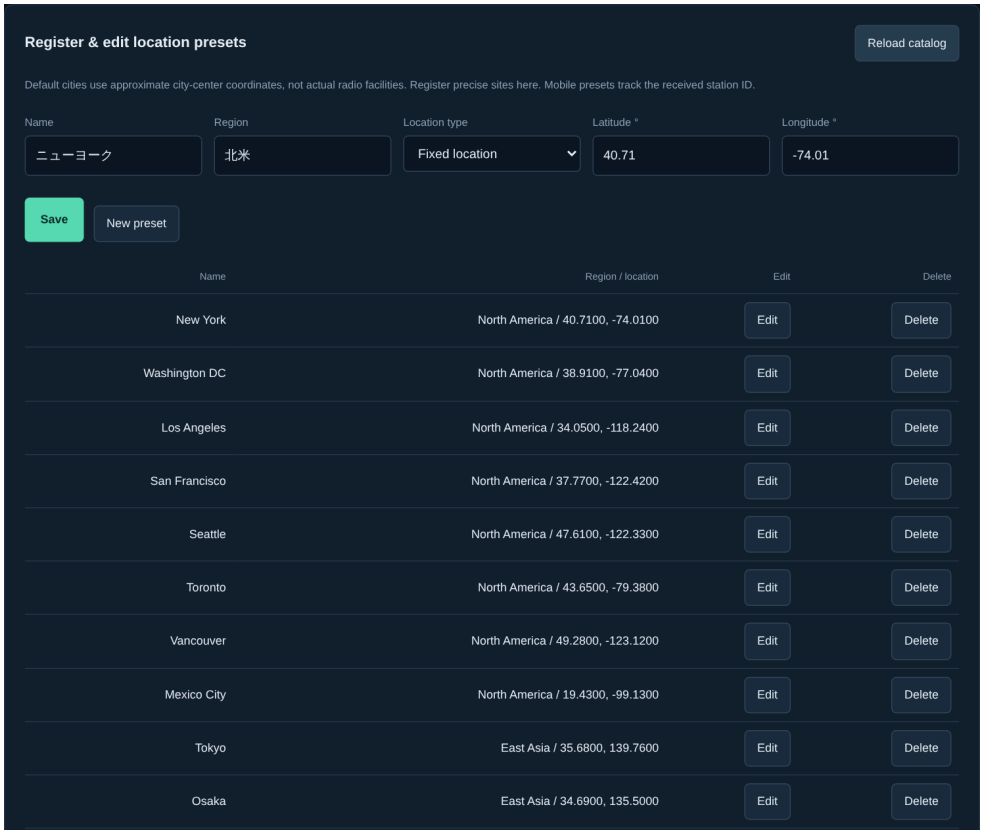


Figure 30 Location preset editing screen (top of the list)

1. “New preset” clears the edit fields; enter the name and region.

2. For a fixed location, enter the latitude and longitude. The latitude for forecasting is −89 to 89°.

3. To choose “Received mobile station”, select the received station and register the position reference.

4. Press “Save” and use the location in the location selection on the right. Manage existing items with edit and delete in the list.

> The initial cities are approximate coordinates. To use actual station positions, register your own locations. If simultaneous edits conflict, reload the list and edit again. Up to 200 locations can be registered.

20 Antenna presets and gain tables

Register the model, dimensions, installation conditions and gain correction in “Antenna preset”. The figure shows a horizontal dipole with a total length of 20 m.



Figure 31 Antenna registration fields

1. With “New preset”, choose the name and base model. There are omnidirectional models (whip, isotropic, vertical monopole, vertical dipole) and directional models (horizontal dipole, horizontal log-periodic, inverted L, sloping long wire, gain table).

2. For models with dimensions, enter the total length in m, the height above ground / feed point height in m, and the ground constants. For directional models, choose the main radiation direction from “Fixed bearing” or “Toward the other station”.

3. Enter the gain correction, confirm its basis and that nothing is counted twice, and save.

4. Assign the antenna to TX or RX in the antenna selection on the right and recalculate.

Gain tables (measured or analysed patterns)

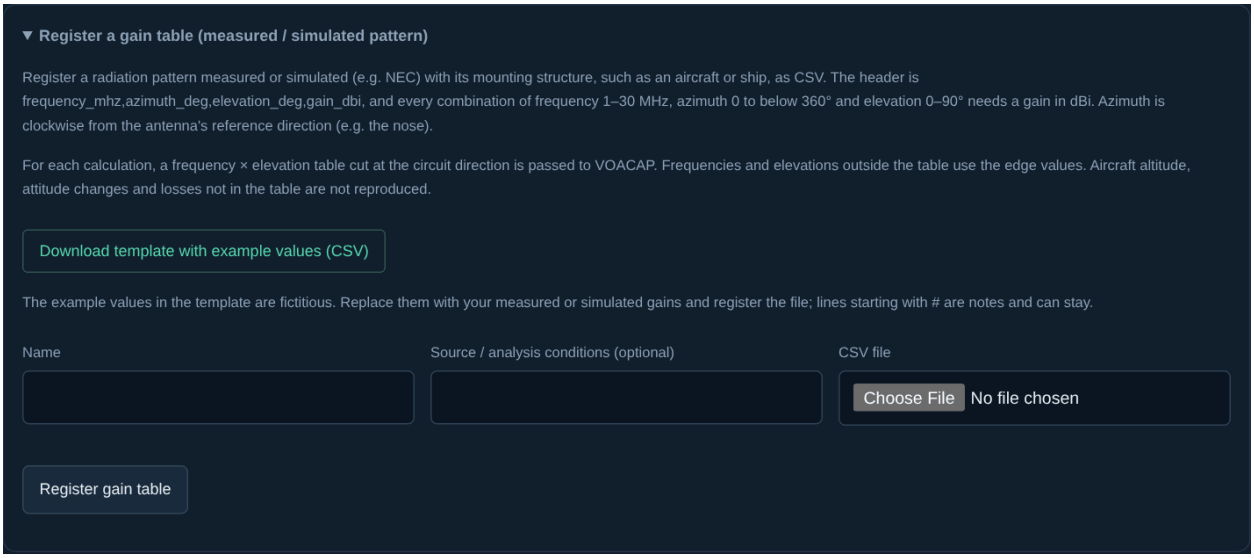


Figure 32 Registering a gain table and the CSV template

1. Save the template with “Download template with example values (CSV)”.

2. With the header frequency_mhz,azimuth_deg,elevation_deg,gain_dbi, replace the values with gains (dBi) that were measured or analysed with NEC or similar.

3. Specify the name, source / analysis conditions and the CSV file, and press “Register gain table”.
4. Choose “Gain table” as the base model of the antenna, assign the registered gain table and save.

Distinguish physical length, electrical length and installation height. Details such as the airframe or hull, matching units, SWR and radials are not reproduced automatically. For frequencies and elevation angles outside the range of the gain table, the edge values are used. Up to 200 antennas can be registered.

21 Updating space weather and external data

Fetch external data under “Forecast & space weather”. Distinguish the values from external providers from the SSN used in the calculation.

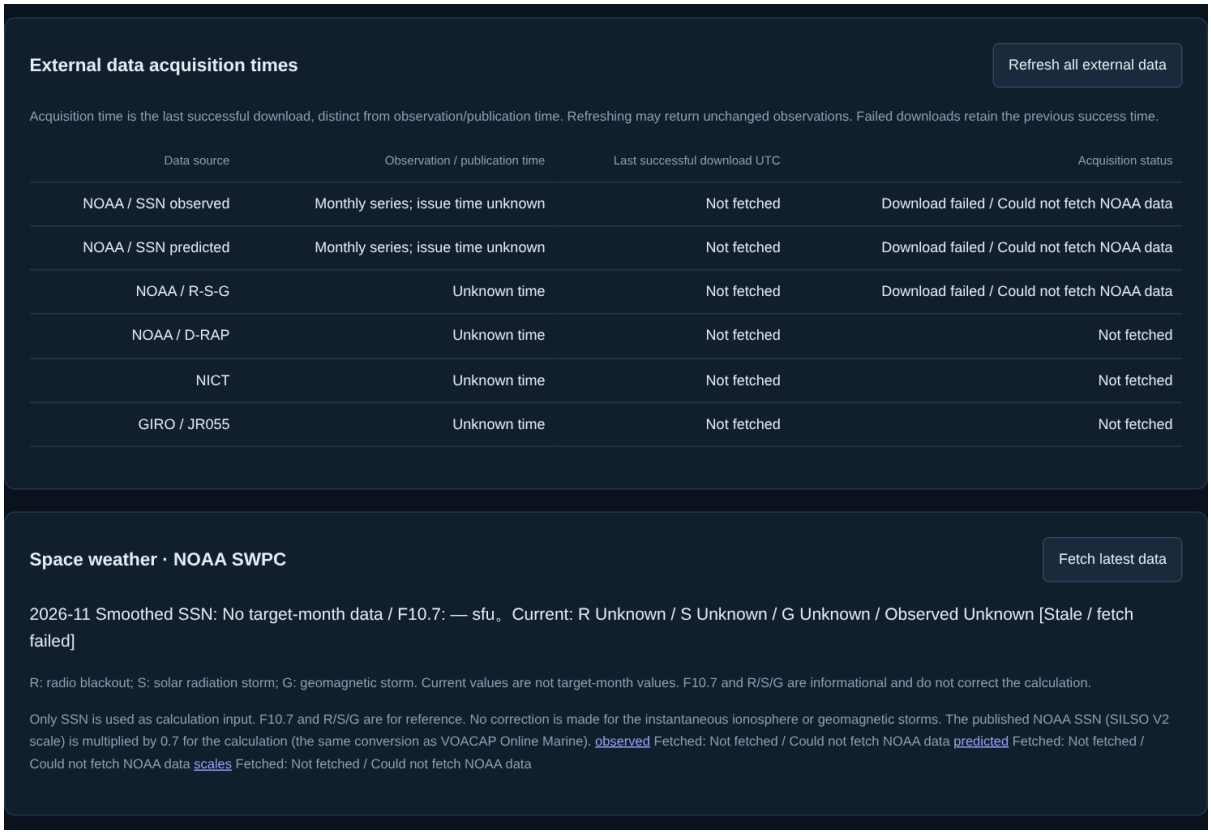


Figure 33 Fetch times of external data and space weather (example before download)

1. “Refresh all external data” updates NOAA, Japanese observations and other data at once. The individual fetch buttons can also be used.

2. Check the observation / issue time and the “last successful fetch time” separately. Times are in UTC.

3. If a provider updates late, the observed values stay old even when the fetch time is new. On failure, the previous successful fetch time is kept.

4. To use the NOAA smoothed SSN in the calculation, switch the SSN source to NOAA and calculate.

> F10.7, R (radio blackouts), S (solar radiation storms) and G (geomagnetic storms) are shown for reference only. They are not used for any additional propagation correction. The automatic NOAA SSN requires the target month to match and does not silently substitute daily values or adjacent months.

Ionospheric observations and absorption

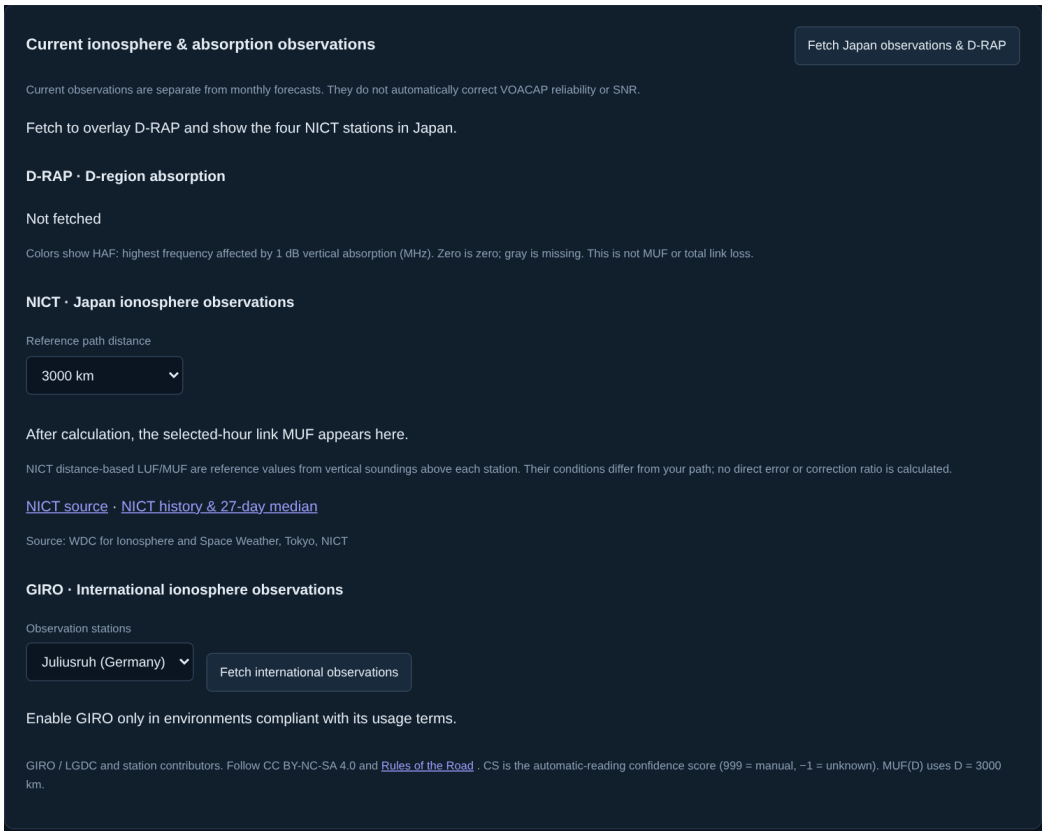


Figure 34 D-RAP, NICT and GIRO controls

1. “Fetch Japan observations & D-RAP” fetches the four Japanese observation stations and absorption information.

2. “D-RAP absorption” on the map shows the grid. The colour is the HAF, not the loss of the whole link or the MUF.

3. For NICT, choose a reference propagation distance and check LUF/MUF. Also check the observation time and missing data.

4. For GIRO, in an environment where it has been enabled under its conditions, select an observation station and press “Fetch international observations”. The effective SSN is calculated from the foF2 fetched here.

> GIRO is disabled by default. Check its terms of use, obtain the necessary permission, then set HF_GIRO_ENABLED=1 in .env and restart. Observed values do not automatically correct the VOACAP REL/SNR.

22 Preparing the gateway and logging in

Receiving external positions and providing them to the main app is handled by an independent gateway. No token settings are needed for normal calculations between fixed locations. It is not available in the public version.

Setting	Purpose and conditions
HF_POSITION_INGEST_TOKEN	External transmitter → gateway. At least 32 characters.
HF_POSITION_GATEWAY_TOKEN	Main app → gateway. At least 32 characters and different from the ingest token.
HF_GATEWAY_UI_PASSWORD	Login to the web administration. At least 16 characters and different from both tokens.
HF_POSITION_MAX_AGE_SECONDS	Expiry from the observation time. Default 300 seconds, 10 to 86400 seconds.

Set the above in .env or similar directly under the project, and apply with docker compose up -d --build. Do not include secret values in Git, screen documentation or sending URLs.

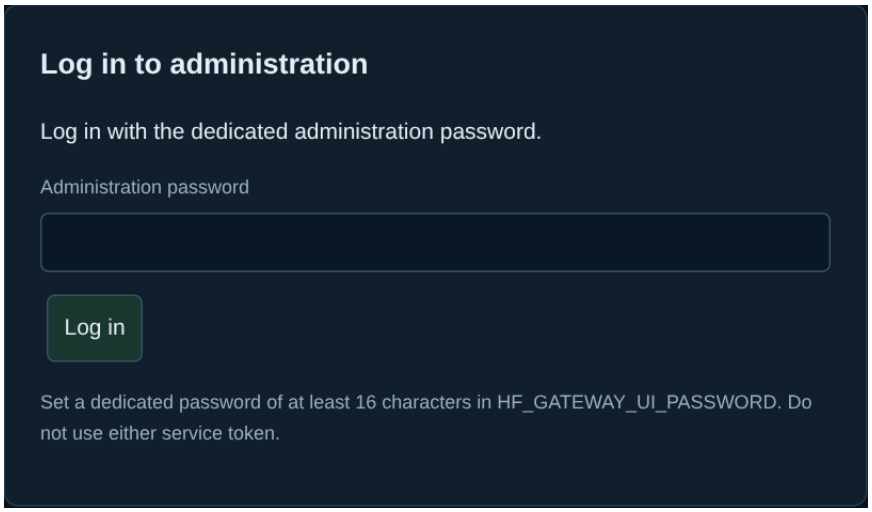


Figure 35 Gateway login screen

1. Open http://localhost:18083 and log in with the administration password.

2. On the same machine, open http://localhost:18082/api/positions and check for a normal response with schema_version 1.

3. After an external transmission, check that the received station also appears in the main app.

healthy and /api/health only check that the DB is alive and do not guarantee successful token authentication. Between containers, position-gateway:8001 is used; this is separate from changing the published host port.

23 Using the gateway web screen

The administration screen manages the input format and reception status of each station. There is no need to enter the service tokens in the browser.

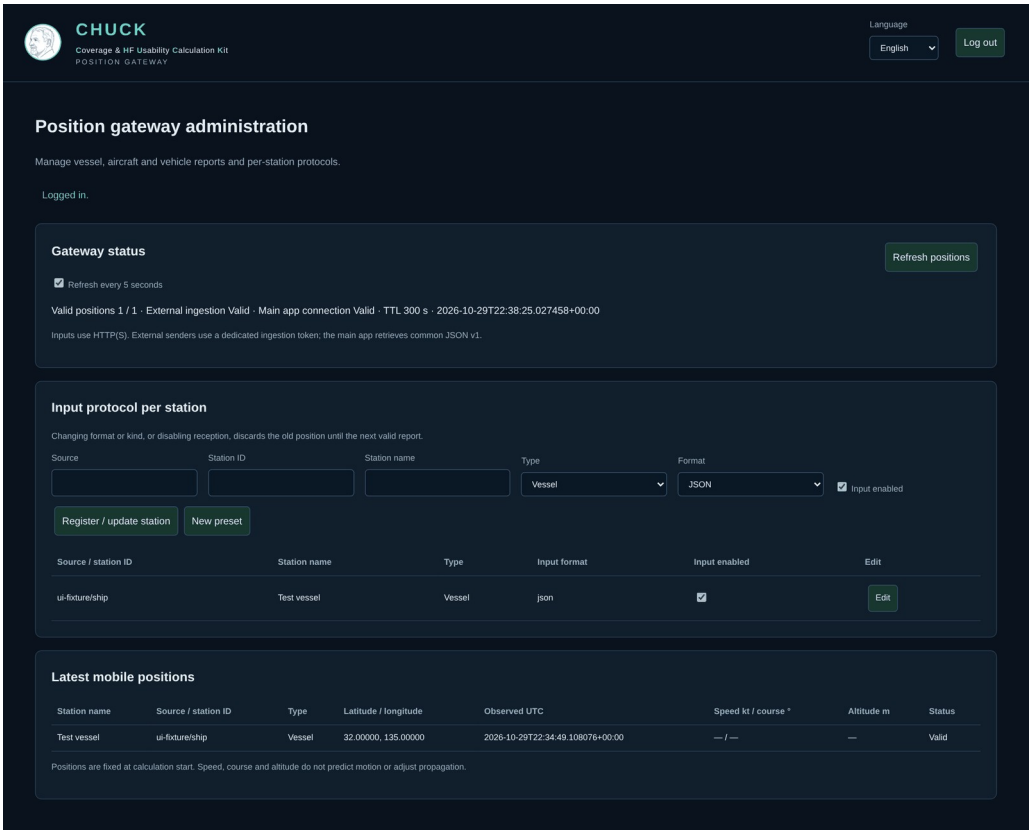


Figure 36 Status, station settings and latest positions on the administration screen

1. Enter the sender source and station ID, the station name, and the type: ship, aircraft or vehicle.
2. Choose the format from JSON, GeoJSON and NMEA 0183 RMC, check “Input enabled” and save.
3. Use “Edit” in the list to load an existing station. Changing the format or type, or disabling the station, discards the old position.
4. Check the latest position, observation time and expiry. The 5-second refresh can be stopped.
5. Log out when you finish. A session expires after 1 hour and is also invalidated by a restart.

NMEA stations must be registered before sending. JSON/GeoJSON stations can be registered automatically by their first valid report. Use the one selected format for each station.

24 Using mobile stations in forecasts

In “Mobile stations & input” in the main app, check the reception status and assign stations to TX/RX from the location presets.

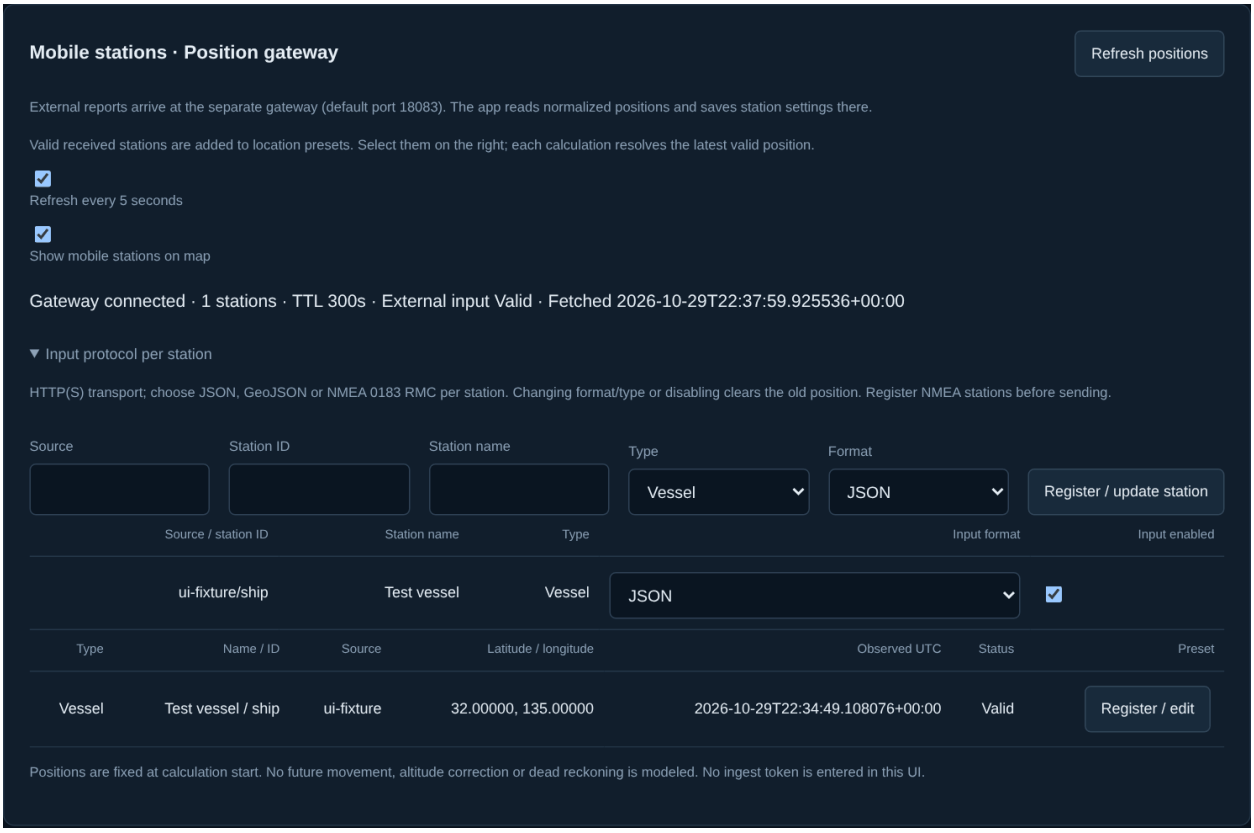


Figure 37 Position list and protocol settings in the main app

1. External transmitters send positions to port 18083 of the gateway. Position POSTs to the former main app port 18082 have been discontinued.

2. Check valid positions with “Refresh positions”. Valid received stations are registered automatically as location presets. “Show mobile stations on map” also shows them on the map.

3. On the right, choose the location preset of the mobile station (marked [M]) and check that “Tracks latest position” is shown.

4. Run the forecast. The position and observation time used are saved with the results. Returning to a fixed location ends the tracking.

> JSON coordinates are lat/latitude and lon/longitude, and GeoJSON coordinates are in the order longitude, latitude. Times use ISO format with a time zone; for NMEA, send valid RMC sentences with a checksum.
>
> Positions older than the default 300 seconds are not used in calculations. The position is fixed at the start of the calculation; the 24-hour movement track and propagation corrections for aircraft altitude are not calculated.

25 Setting the AI, model and API key

Open “Antenna preset” → “Read catalogs/specifications with AI” → “AI, model and API key settings”.

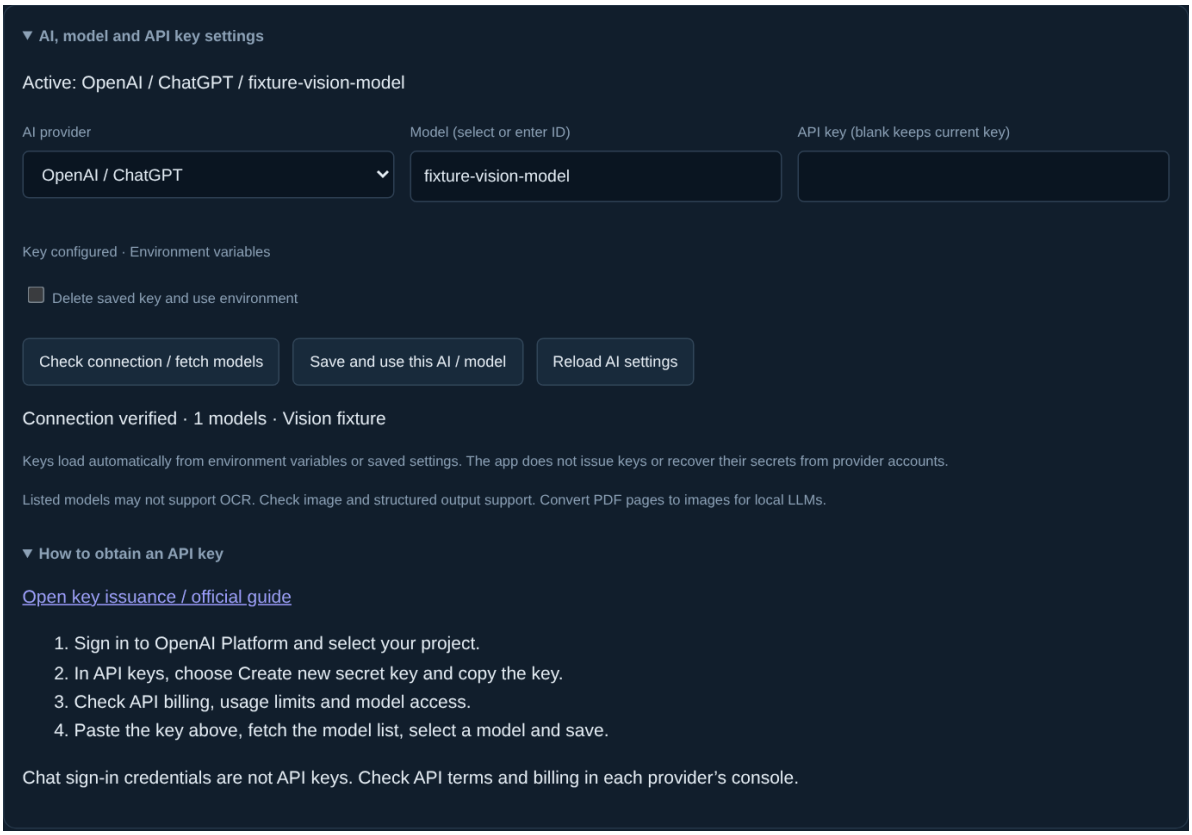


Figure 38 AI settings screen (model names are illustrative samples)

1. Choose the AI to use from OpenAI/ChatGPT, Claude, Gemini and Local LLM.

2. Open “How to obtain an API key” and enter a key issued in each provider’s API console.

3. Press “Check connection / fetch models” and choose a model that supports image input and structured output. You can also enter the ID directly.

4. Press “Save and use this AI / model”. Selecting an AI alone does not apply it to OCR.

Address	Key issuance or address
OpenAI	https://platform.openai.com/api-keys
Claude	https://platform.claude.com/settings/keys
Gemini	https://aistudio.google.com/api-keys
Ollama example	http://host.docker.internal:11434
OpenAI-compatible local example	http://host.docker.internal:1234/v1

Automatic key retrieval means reading from saved settings or environment variables. Keys are not issued automatically by providers, and secret values are not recovered. Check API billing and usage limits separately from chat subscriptions such as ChatGPT. Saved keys are not stored encrypted.

26 Reading and registering antenna documents

Importing creates candidates. AI reading alone does not register anything as an antenna preset.

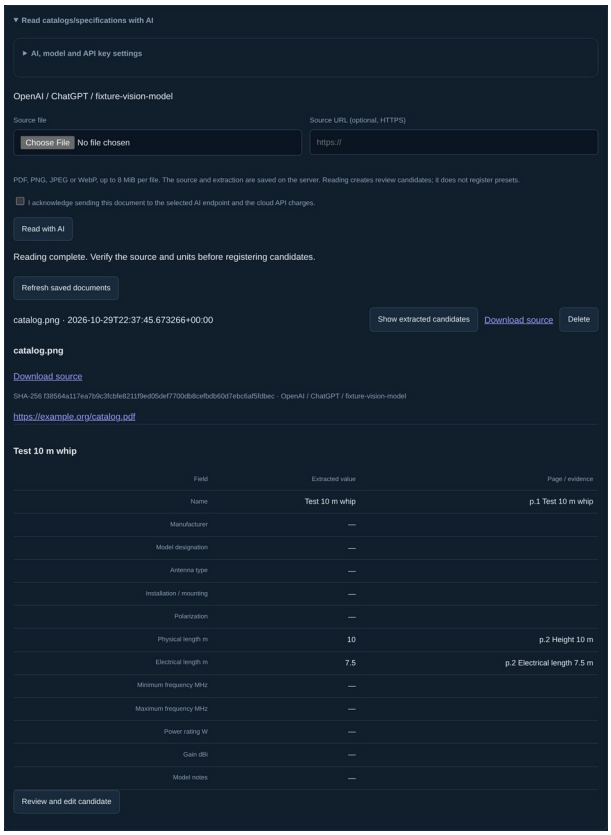


Figure 39 Read candidates and source pages (read values are illustrative samples)

1. Choose the document file: PDF, PNG, JPEG or WebP, up to 8 MiB per file. A local LLM accepts images only, so PDFs are converted to page images.

2. If needed, record the HTTPS source URL. This is not a feature that fetches documents from the URL automatically.

3. Check where the document is sent and the API charges, tick the checkbox and press “Read with AI”.

4. Check the candidate values, units and source pages, and press “Review and edit candidate”.

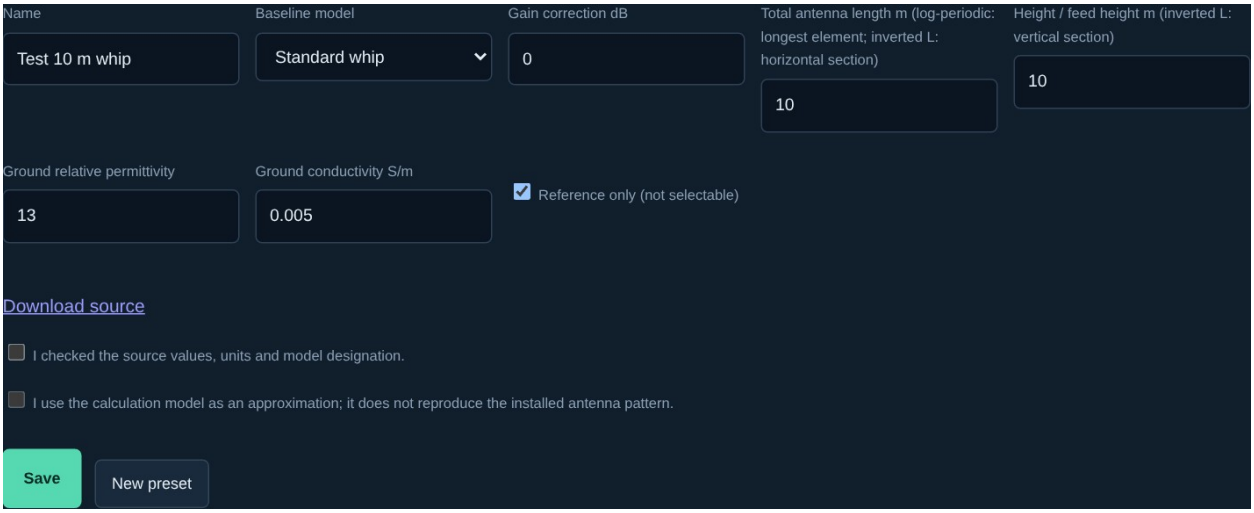


Figure 40 Fields to confirm the source document and the approximate use

1. Open the document for the model with “Download source” and compare the values, units and source pages.

2. Correct the name, base model, gain correction, dimensions and installation conditions, and tick the source confirmation.

3. To keep it for reference, leave “Reference only” on and save. To use it in calculations, set an appropriate model and dimensions, clear the reference flag, confirm that it is used as an approximation and save.

> Do not confuse physical total length with electrical length, height above ground, dBd with dBi, or transmit output with permissible power. Do not fill unknown values with guesses; save them as reference material. Source documents and read results are stored on the server, up to 50 documents. A document cannot be deleted while a preset refers to it.

27 Calculation history and scheduled recalculation

“Calculation history” shows saved conditions and results again. The history lists the latest 100 entries; the number of saved entries itself is unlimited.

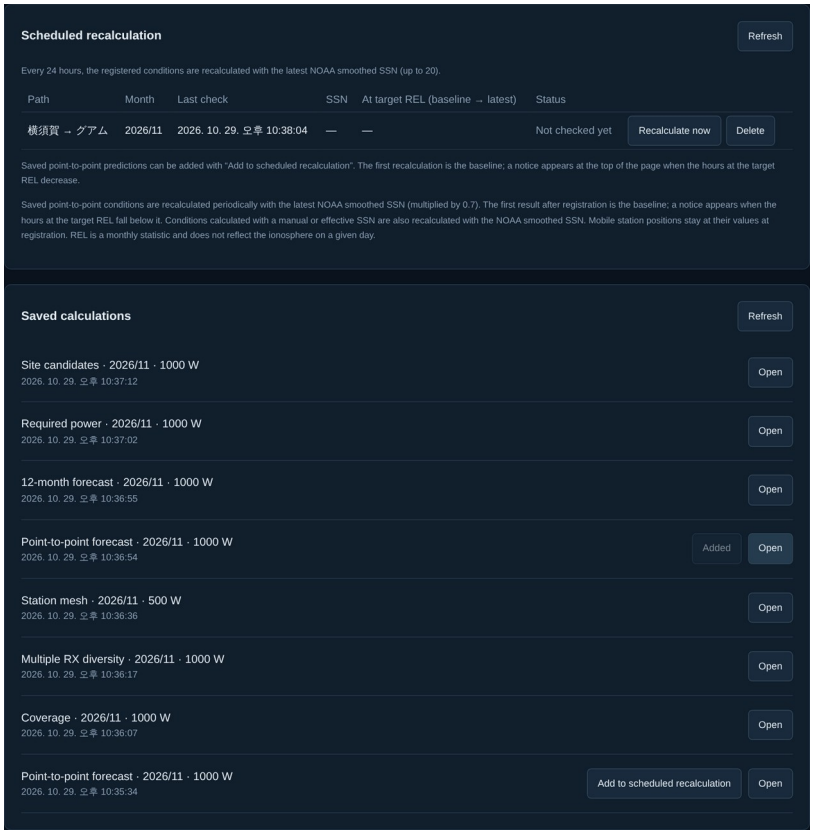


Figure 41 Scheduled recalculations and saved calculations

1. “Refresh” fetches the history list; press “Open” for the entry you want. Point-to-point, diversity, mesh, coverage, 12-month forecast, required power and site candidate results can be shown again.

2. Check the conditions and results at the time of calculation. The positions of mobile stations at that time are also kept in the results.

3. With “Add to scheduled recalculation” for a point-to-point forecast, the server recalculates every 24 hours by default with NOAA’s latest smoothed SSN. The first recalculation after registration is the baseline.

4. If the number of hours at or above the target REL falls below the baseline, the status becomes “Decreased” and a notice appears at the top of the screen. “Mark as reviewed” clears the notice; “Recalculate now” and “Delete” are also available.

Saving method	What is saved
Save settings	Saves the input conditions and the TX/RX selection in that browser. It is not a backup of the history or API keys.
Calculation history	Saves the conditions and results at the time of calculation on the server (in the browser for the public version).

CSV, JSON, report	For taking results elsewhere. Not a complete backup of locations, antennas, AI settings or source documents.
Persistent volumes	To preserve everything including settings and source documents, back up hf-data and gateway-data separately while stopped.

If no fresh NOAA value can be fetched, the status is “Download failed” and the baseline and hours are not changed. Up to 20 scheduled recalculations are possible, and the interval can be changed with HF_WATCH_INTERVAL_HOURS (0 to stop). In the public version only the administrator can use this feature.

28 What’s new

With “What’s new” at the top right of the screen, you can check the history of features added since the initial release. Features added since you last opened it are marked NEW, and their number is shown on the button.

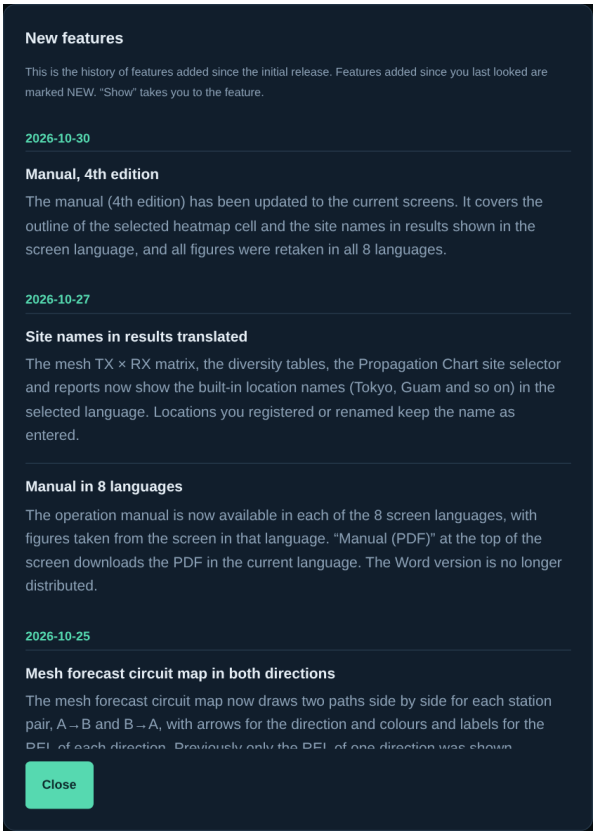


Figure 42 What’s new

1. Press “Show” to go to the screen of that feature.
2. “Close” closes the list. When there are new features, a notice also appears at the top of the screen.

29 Troubleshooting

Check the error shown and the logs, and narrow down the cause from the related settings.

Symptom	Check and fix
Compose YAML error	After git pull --ff-only, run docker compose config --quiet. If you changed the file locally, check the indentation of environment, extra_hosts and so on.
The screen does not open	Check that Docker is running, docker compose ps, the logs and conflicts on port 18082. If you changed the port, change the URL too.
The engine does not become ready	Check for build failures and the logs. Do not judge the engine to be healthy from the gateway being healthy.
Calculating, or HTTP 409	Wait for the other calculation to finish. For a conflict in saved settings, reload and edit again.
Cannot calculate in the public version	Check whether the calculation limit has been reached. Mesh, 12-month, power and site candidate calculations use several calculations at once.
The report does not open	Allow pop-ups for this page in your browser.
A mobile station cannot be calculated	Check the latest position, observation time, the 300-second expiry, whether the station is enabled and latitude ±89°.
HTTP 503 on the position link	Check that the main app token is the same in both services, at least 32 characters and different from the ingest token.
HTTP 401 on position input	Set the ingest token on the sending side. The main app token or the administration cookie cannot be used instead.
NMEA is not received	Register the NMEA station with the same source/id first. Check the RMC, checksum, status and date and time.
NOAA fetch failed	Check the HTTPS connection, the target year and month and the cache state. If needed, return to manual SSN.
AI reading is not available	Check that the AI and model are saved, the key, the API usage limit, support for images and structured output, and that the file is within 8 MiB.
Cannot connect to the local LLM	Check the host address (not localhost inside the container), that it is listening, the firewall and the API format.
A reference antenna cannot be selected	It is for reference only. It cannot be used in calculations until a well-founded model and dimensions and the necessary confirmations are in place.

Checking logs: docker compose logs --tail=100 -f. When asking for help, record the error, the time, the steps to reproduce and the version used, and remove API keys, tokens and confidential documents.

30 Checks before operation and terms of use

Before using calculation results or sharing them externally, check the input conditions and the terms of use of the data.

Checks before calculating

1. Check the TX/RX locations and antennas, the year and month, SSN, power and noise.

2. Distinguish UTC from Japan time, and check the required SNR, target REL and that gains and losses are not counted twice.

3. For mobile stations, check the reception time and expiry; for external observations, check the observation time and the successful fetch time.

4. Recalculate after changes, and check that the conditions in the results and reports match the intended settings.

Scope of the forecast

VOACAP is a monthly statistical model. It does not fully reproduce instantaneous ionospheric variations, magnetic storms, terrain or actual installation conditions. Recommended frequencies, operating frequency plans and reports do not determine frequency assignments or operating permissions.

External provision and redistribution

New code in the main app is under MIT, but the bundled software and external data have separate terms. GIRO has terms such as non-commercial research and sharing. Do not treat third-party data via NOAA or NICT data as free for unrestricted use; check the attribution and terms.

See “External data terms & sources” at the bottom of the screen. Do not publish source documents, OCR quotations, mobile station positions, API keys or saved settings as they are. Redistribute the software by following the prescribed bundle procedure.

Network exposure

The main app assumes localhost and has no individual logins. For LAN use, set HF_BIND and HF_GATEWAY_BIND separately and put the necessary authentication, HTTPS and access control in front. For publishing on the internet, use the public version setup (docs/PUBLIC_DEPLOYMENT.md).

Detailed documents

Repository https://github.com/UeEmon/CHUCK-public-
README.md / docs/MESH.md / docs/ANNUAL.md / docs/POWER.md / docs/SITES.md
docs/REPORT.md / docs/FREQUENCY_PLAN.md / docs/WATCH.md / docs/SPACE_WEATHER.md
docs/GATEWAY_PROTOCOL.md / docs/POSITION_INTERFACE.md / docs/AI_SETTINGS.md /
docs/ANTENNA_OCR.md
DATA_LICENSES.md / SOFTWARE_LICENSES.md / docs/REDISTRIBUTION.md

The screens and procedures in this manual are based on the screens as of the date on the cover. After an update, get the latest version from “Manual (PDF)” in the app, and also check the README and the detailed documents.